

Digital Design (SET111)

L12

FSM and Counter

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- ☐ **Finite State Machine**
- ☐ **Asynchronous Counter**
- ☐ **Synchronous Counter**
- ☐ **Design of Synchronous Counters**
- ☐ **Exersice_01**
- ☐ **Exersice_02**
- ☐ **Exersice_03**
- ☐ **Exersice_04**

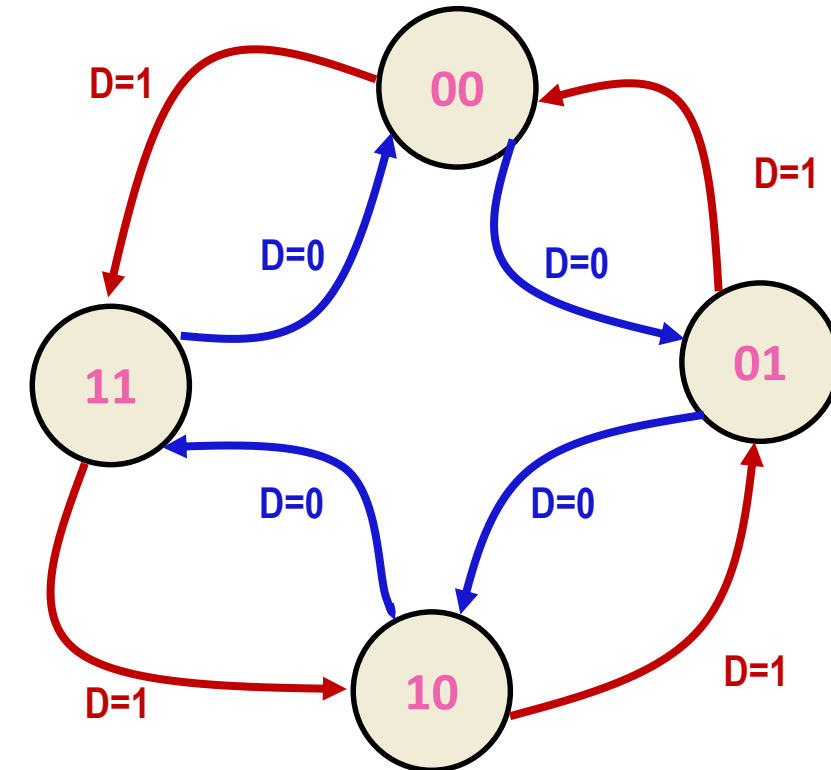
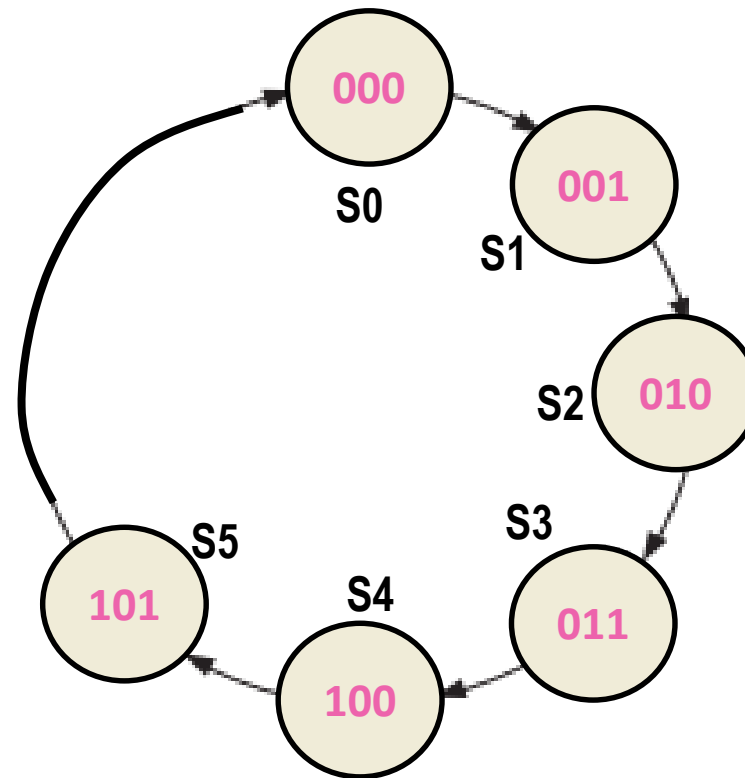
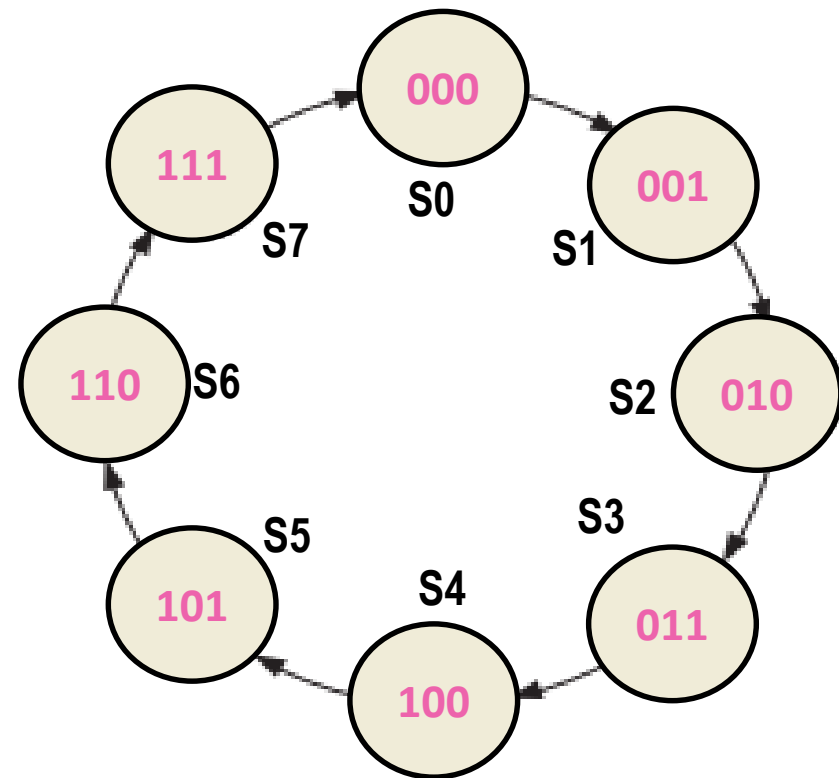


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- ☐ **Exersice_03**
- ☐ **Exersice_04**

Finite State Machine



- ❑ A **state machine** is a sequential circuit having a limited (**finite**) number of states occurring in a prescribed **order**, so it is called Finite State Machine (**FSM**).
- ❑ A **counter** is an example of a state machine, the number of states is called the **modulus**
- ❑ Two basic types of counters are the **Asynchronous** and **Synchronous** counter.





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The clock input of an asynchronous counter is always connected only to the LSB flip-flop

- SET111, Digital Design, Dr. Ahmed Reda**



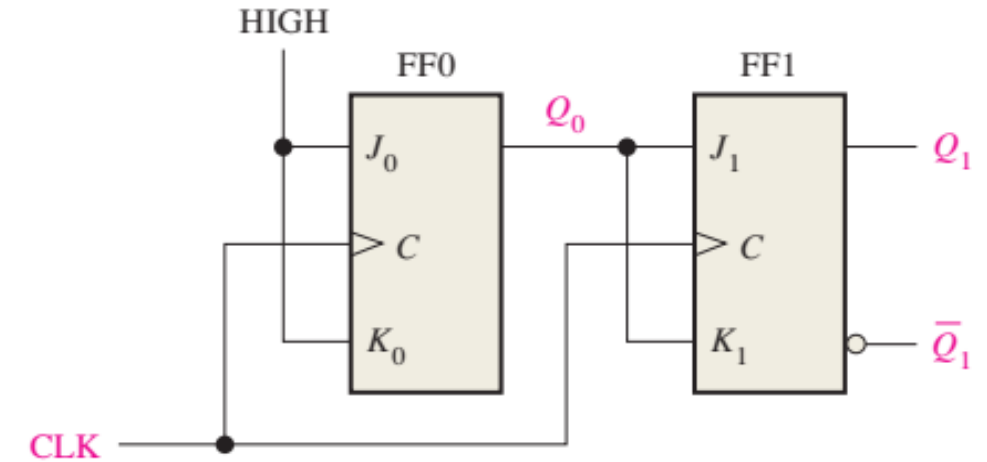
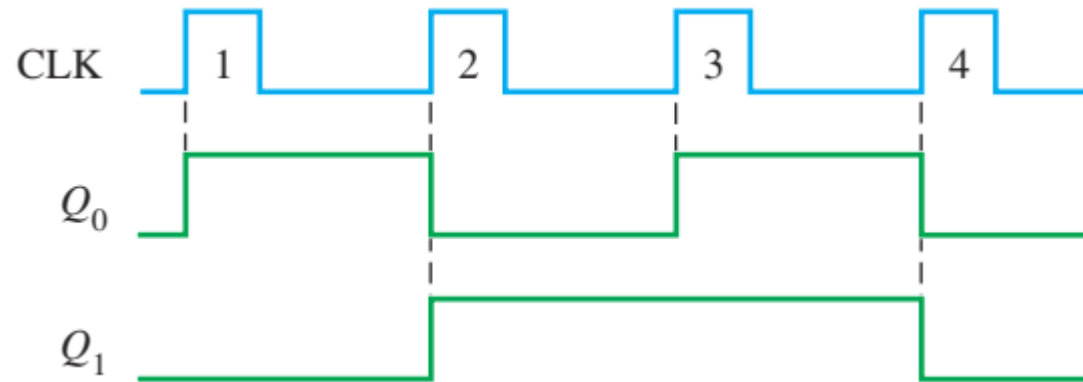
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- ☐ Exercice_03

-
- The diagram shows two J-K flip-flops, FF0 and FF1, connected in a 2-bit counter configuration. FF0's J_0 input is connected to a HIGH signal, and its K_0 input is connected to the CLK signal. The output Q_0 of FF0 is connected to the J_1 input of FF1. FF1's K_1 input is connected to the CLK signal. Both flip-flops have their clock inputs (C) connected to the CLK signal. The outputs of FF1 are Q_1 and $\overline{Q_1}$.

The diagram shows a 2-bit counter implemented with two D flip-flops, labeled D_0 and D_1 . Both flip-flops have a clock input C connected to a common CLK signal. The output of D_0 is Q_0 . This output is connected to the input of a 2-input XOR gate. The other input of the XOR gate is the output of D_1 , which is Q_1 . The output of the XOR gate is connected back to the input of D_1 . The output of D_1 is Q_1 .

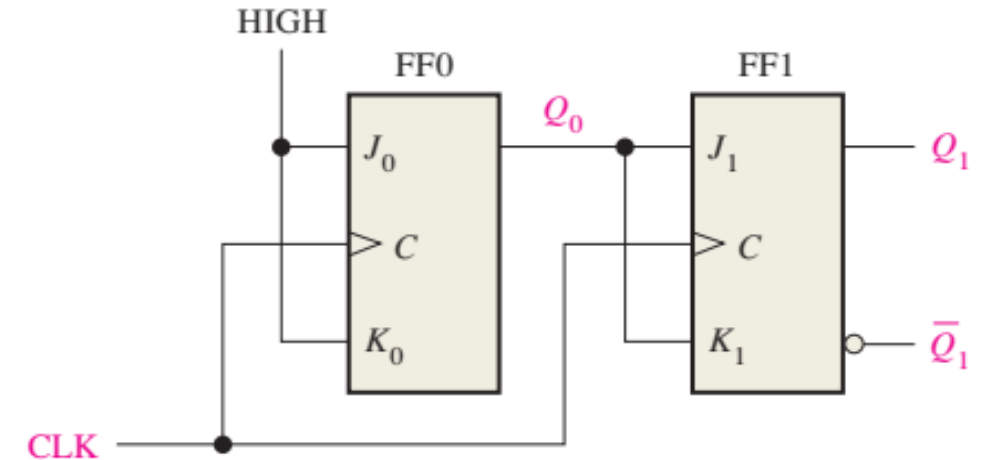
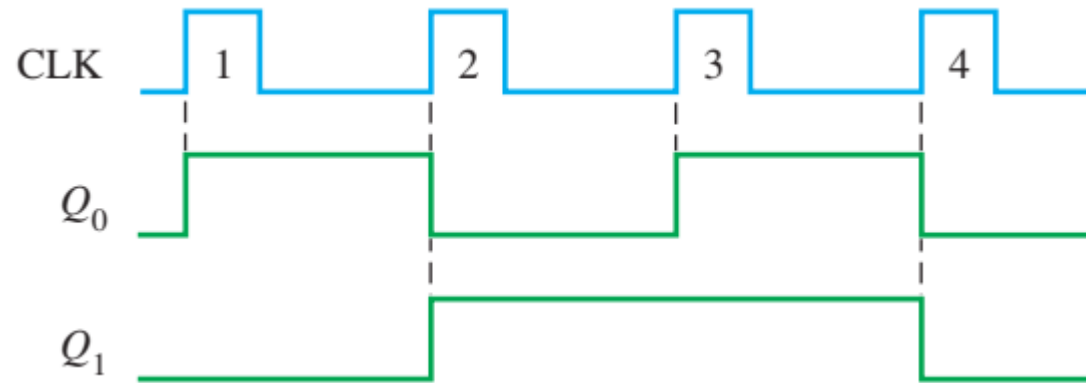
- ❑ J-K flip-flops are used to illustrate most synchronous counters.
- ❑ D flip-flops can also be used but generally require **more logic** because of having no direct toggle or no-change states.

Synchronous Counter (2/4)

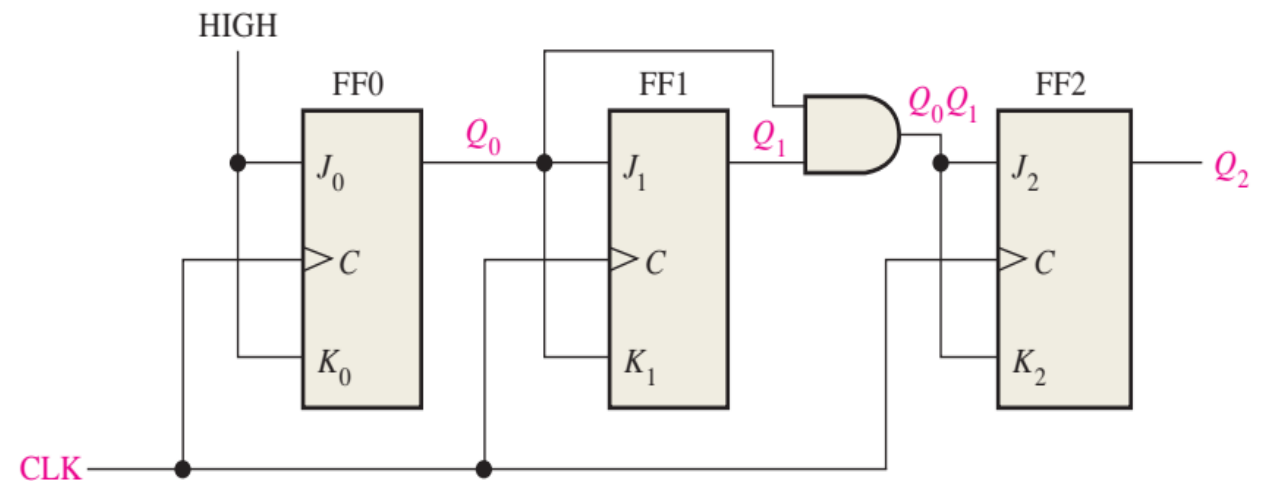
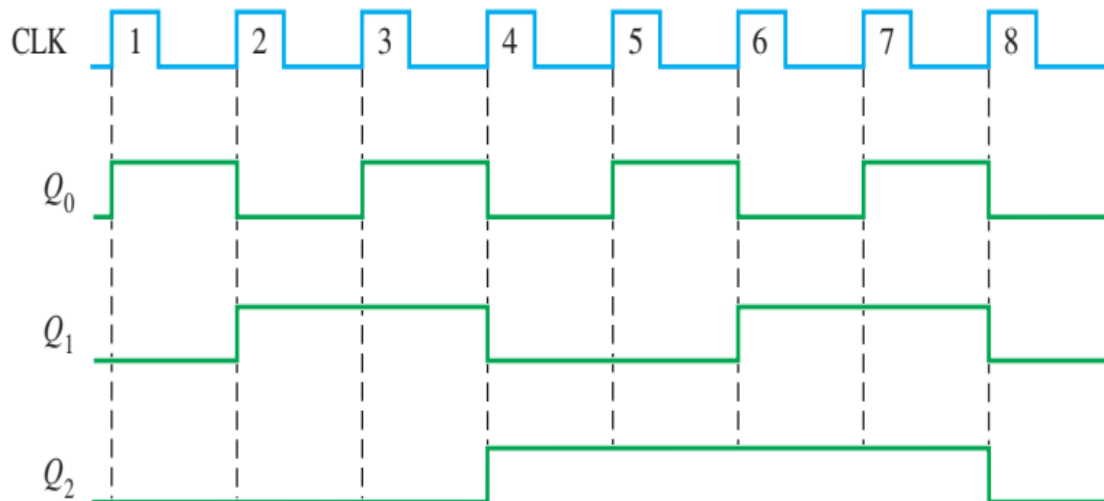


A 2-Bit Synchronous Binary Counter using
(JK Flip Flop)

Synchronous Counter (3/4)

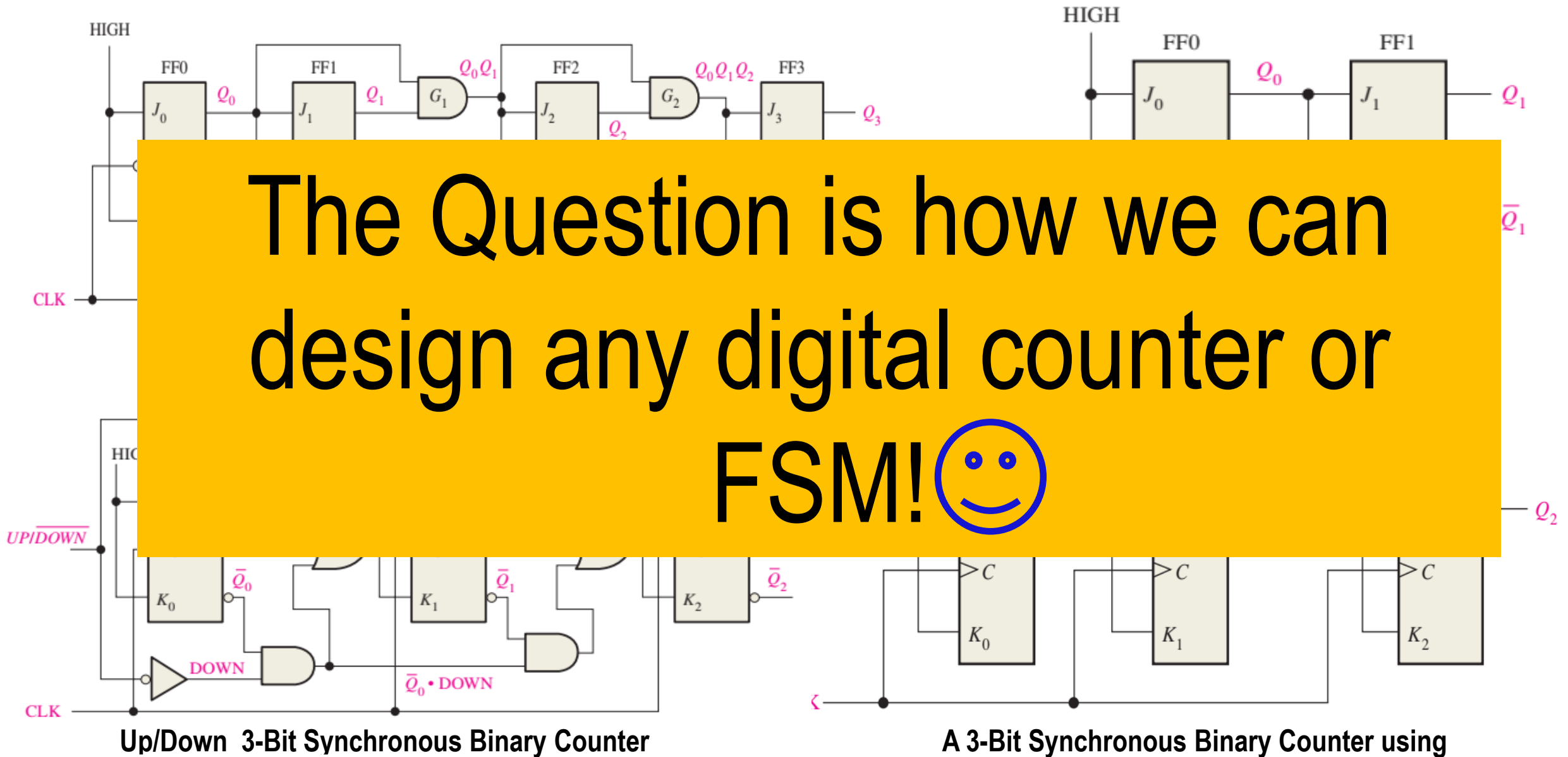


A 2-Bit Synchronous Binary Counter using (JK Flip Flop)



A 3-Bit Synchronous Binary Counter using (JK Flip Flop)

Synchronous Counter (4/4)





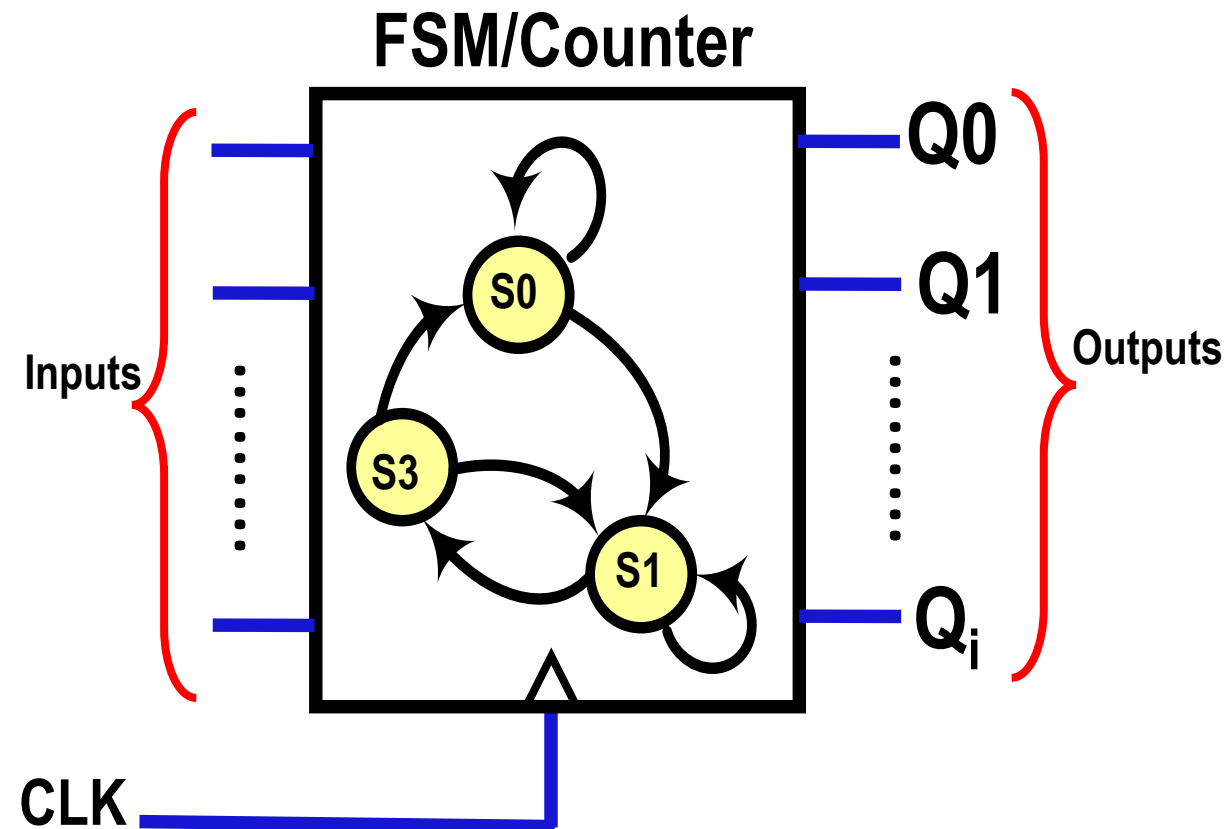
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- ☐ Exercise_04

Design of Synchronous Counters



❑ You have to execute **Five** steps to design a **counter** Or even any finite state machine (**FSM**).

- ❑ **Step 1:** State Diagram
- ❑ **Step 2:** Next-State Table
- ❑ **Step 3:** Flip-Flop Transition Table
- ❑ **Step 4:** K-Maps
- ❑ **Step 5:** Counter Implementation





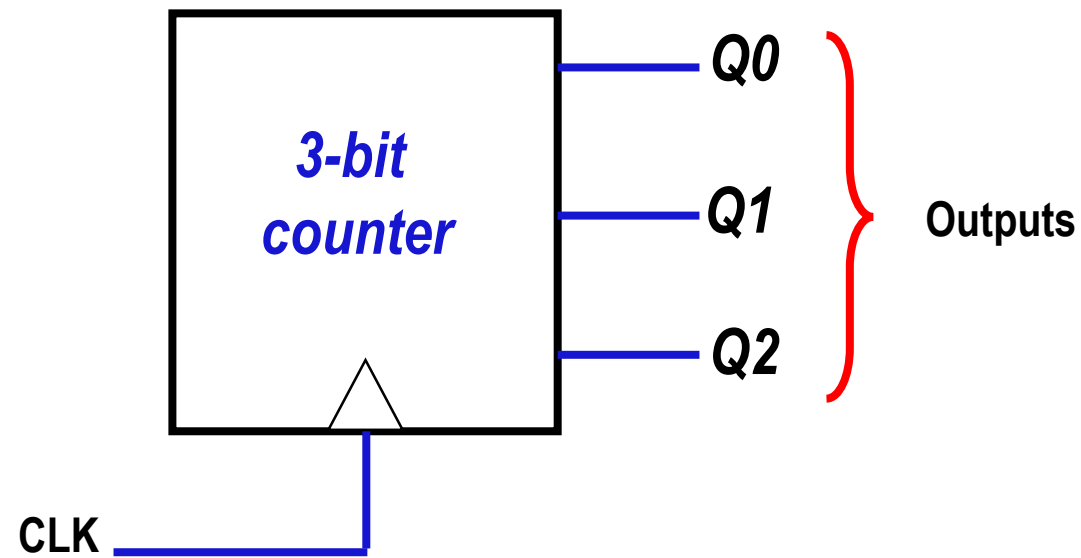
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Exercise_01



□ Design a 3-bit **synchronous** up counter using JK Flip-Flop.

Ans.

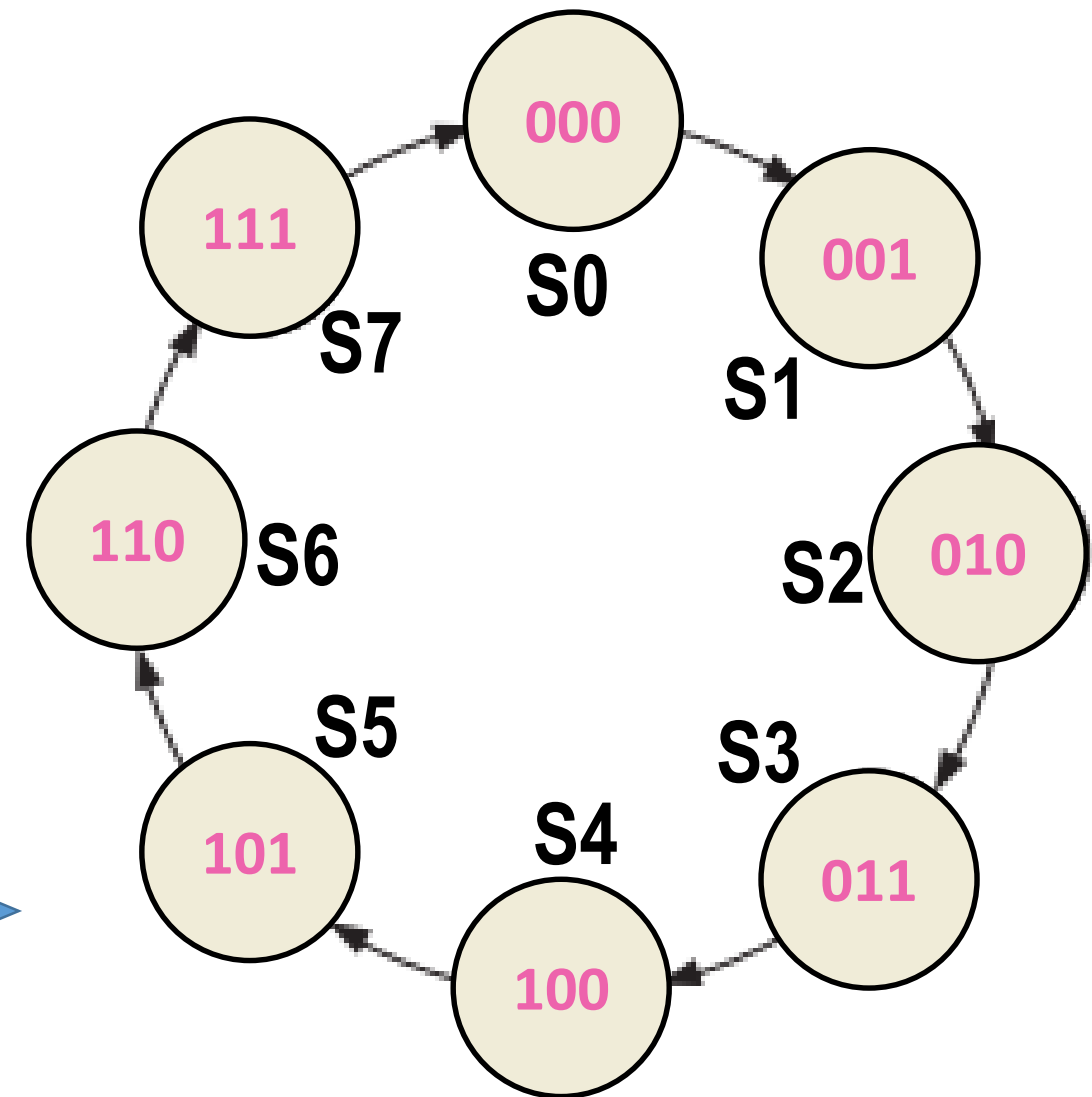
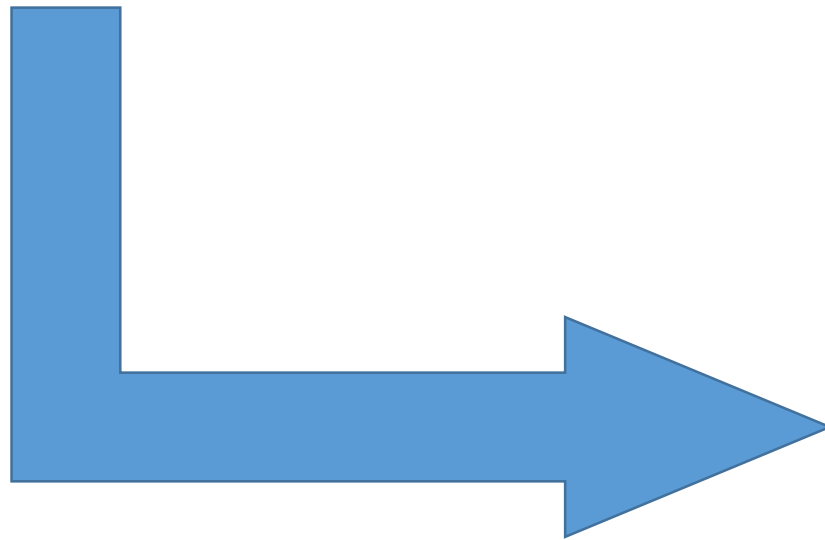


Exersice_01



□ Step 1: State Diagram

3-bit synchronous $\rightarrow 2^3 = 8$ states
Up counter $\rightarrow \{0, 1, 2, 3, 4, 5, 6, 7\}$



State Diagram
3-bit counter

Exercise_01



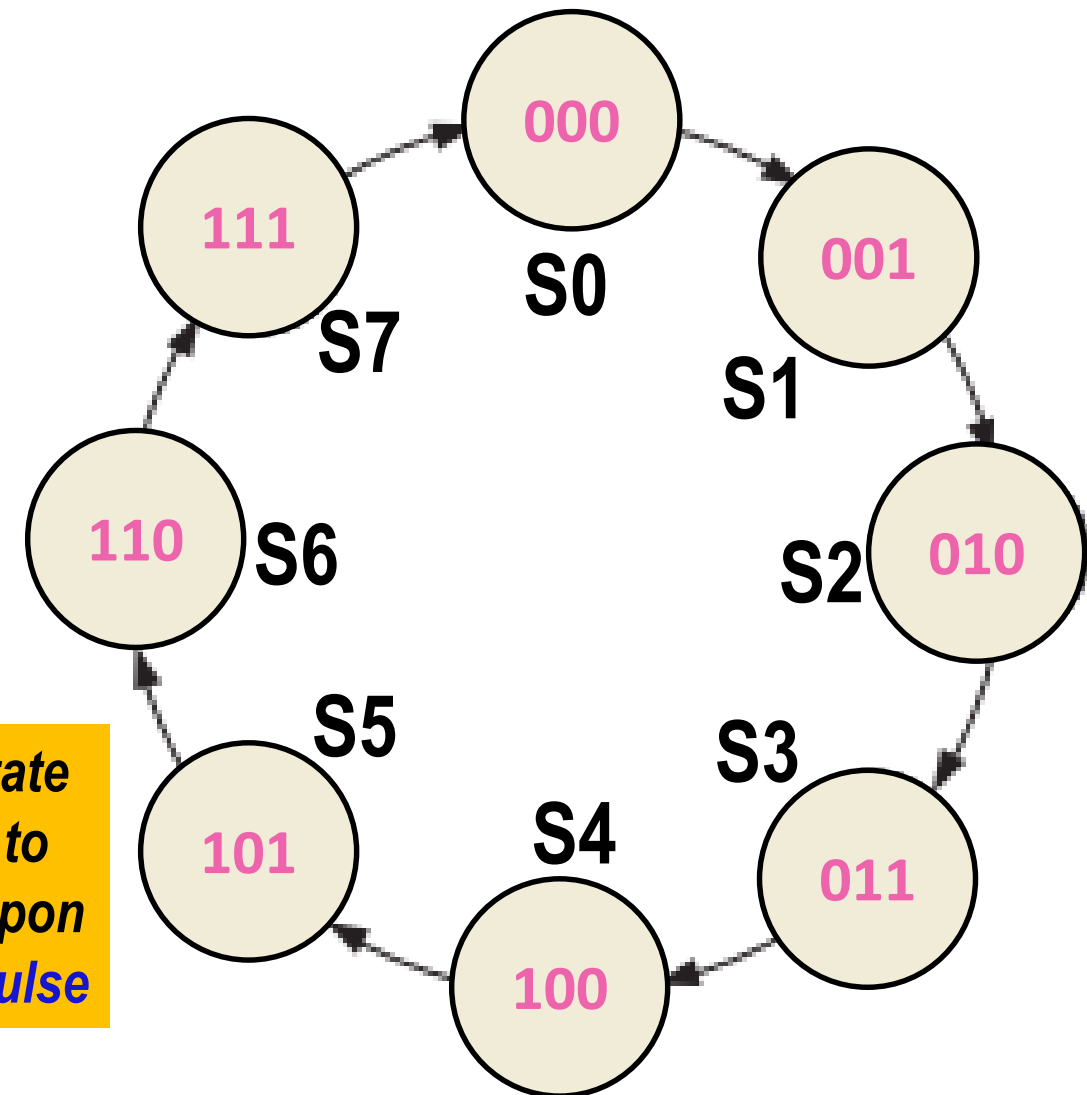
❑ Step 2: Next-State Table

❑ Once the sequential circuit is defined by a state diagram, the **second step** is to derive a next-state table

Next-state table for 3-bit counter

Present State			Next State		
Q2	Q1	Q0	Q2	Q1	Q0

The **next state** is the state that the counter goes to from its present state upon application of a **clock pulse**



Exercise_01



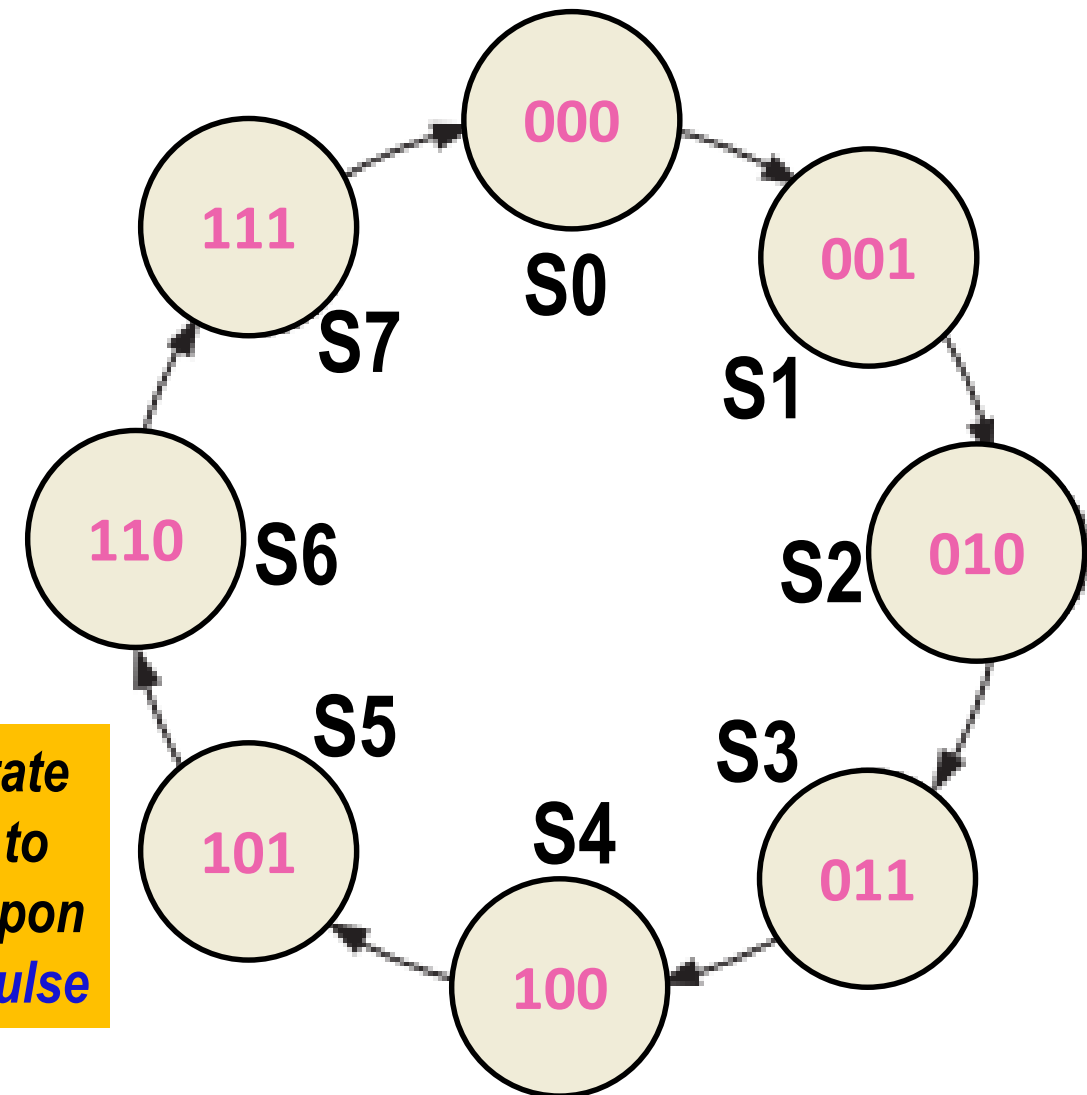
❑ Step 2: Next-State Table

❑ Once the sequential circuit is defined by a state diagram, the **second step** is to derive a next-state table

Next-state table for 3-bit counter

Present State			Next State		
Q2	Q1	Q0	Q2	Q1	Q0
0	0	0	0	0	1
0	0	1	0	1	0
0	1	0	0	1	1
0	1	1	1	0	0
1	0	0	1	0	1
1	0	1	1	1	0
1	1	0	1	1	1
1	1	1	0	0	0

The **next state** is the state that the counter goes to from its present state upon application of a **clock pulse**



❑ Step 3: Flip-Flop Transition Table

❑ For each **output transition**, the J and K **inputs** that will cause the output transition to occur are always the same and defined in table.

Next-state table for 3-bit counter

Present State			Next State		
Q2	Q1	Q0	Q2	Q1	Q0
0	0	0	0	0	1
0	0	1	0	1	0
0	1	0	0	1	1
0	1	1	1	0	0
1	0	0	1	0	1
1	0	1	1	1	0
1	1	0	1	1	1
1	1	1	0	0	0

Transition table for a J-K flip-flop

Output Transistors			Flip-Flop Inputs	
Q_N		Q_{N+1}	J	K
0	→	0	0	X
0	→	1	1	X
1	→	0	X	1
1	→	1	X	0

Q_N : Present State

Q_{N+1} : Next State

X: Don't Care

Exercise_01



❑ Step 3: Flip-Flop Transition Table

❑ # of JK Flip-Flop = # of output of the counter. Here, # of JK Flip-Flop=3.

Next-state table for 3-bit counter

Present State			Next State			Inputs of the JK-Flip Flop					
Q2	Q1	Q0	Q2	Q1	Q0	J2	K2	J1	K1	J0	K0
0	0	0	0	0	1						
0	0	1	0	1	0						
0	1	0	0	1	1						
0	1	1	1	0	0						
1	0	0	1	0	1						
1	0	1	1	1	0						
1	1	0	1	1	1						
1	1	1	0	0	0						

Transition table for a J-K flip-flop

Output Transistors			Flip-Flop Inputs	
Q _N		Q _{N+1}	J	K
0	→	0	0	X
0	→	1	1	X
1	→	0	X	1
1	→	1	X	0

Q_N : Present State

Q_{N+1} : Next State

X: Don't Care

Exersice_01



❑ Step 3: Flip-Flop Transition Table

❑ Map between the each output transition and the inputs of JK Flip-Flop

Next-state table for 3-bit counter

Present State		Next State		Inputs of the JK-Flip Flop		
	Q0		Q0		J0	K0
	0		1			
	1		0			
	0		1			
	1		0			
	0		1			
	1		0			
	0		1			
	1		0			

Transition table for a J-K flip-flop

Output Transistors			Flip-Flop Inputs	
Q_N		Q_{N+1}	J	K
0	→	0	0	X
0	→	1	1	X
1	→	0	X	1
1	→	1	X	0

Q_N : Present State

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X: Don't Care

Exercise_01



Step 3: Flip-Flop Transition Table

Map between the each output transition and the inputs of JK Flip-Flop

Next-state table for 3-bit counter

Present State		Next State		Inputs of the JK-Flip Flop	
	Q0		Q0	J0	K0
	0		1	1	X
	1		0	X	1
	0		1	1	X
	1		0	X	1
	0		1	1	X
	1		0	X	1
	0		1	1	X
	1		0	X	1

Transition table for a J-K flip-flop

Output Transistors			Flip-Flop Inputs	
Q_N		Q_{N+1}	J	K
0	→	0	0	X
0	→	1	1	X
1	→	0	X	1
1	→	1	X	0

Q_N : Present State

Q_{N+1} : Next State

X: Don't Care

Exercise_01



Step 3: Flip-Flop Transition Table

Map between the each output transition and the inputs of JK Flip-Flop

Next-state table for 3-bit counter

Present State			Next State			Inputs of the JK-Flip Flop		
	Q1			Q1			J1	K1
	0			0				
	0			1				
	1			1				
	1			0				
	0			0				
	0			1				
	1			1				
	1			0				

Transition table for a J-K flip-flop

Output Transistors			Flip-Flop Inputs	
Q_N		Q_{N+1}	J	K
0	→	0	0	X
0	→	1	1	X
1	→	0	X	1
1	→	1	X	0

Q_N : Present State

Q_{N+1} : Next State

X: Don't Care

Exersice_01



❑ Step 3: Flip-Flop Transition Table

❑ Map between the each output transition and the inputs of JK Flip-Flop

Next-state table for 3-bit counter

Present State			Next State			Inputs of the JK-Flip Flop		
	Q1			Q1			J1	K1
	0			0			0	X
	0			1			1	X
	1			1			X	0
	1			0			X	1
	0			0			0	X
	0			1			1	X
	1			1			X	0
	1			0			X	1

Transition table for a J-K flip-flop

Output Transistors			Flip-Flop Inputs	
Q_N		Q_{N+1}	J	K
0	→	0	0	X
0	→	1	1	X
1	→	0	X	1
1	→	1	X	0

Q_N : Present State

Q_{N+1} : Next State

X: Don't Care

Exercise_01



❑ Step 3: Flip-Flop Transition Table

❑ Map between the each output transition and the inputs of JK Flip-Flop

Next-state table for 3-bit counter

Present State		Next State		Inputs of the JK-Flip Flop	
Q2		Q2		J2	K2
0		0			
0		0			
0		0			
0		1			
1		1			
1		1			
1		1			
1		0			

Transition table for a J-K flip-flop

Output Transistors			Flip-Flop Inputs	
Q_N		Q_{N+1}	J	K
0	→	0	0	X
0	→	1	1	X
1	→	0	X	1
1	→	1	X	0

Q_N : Present State

Q_{N+1} : Next State

X: Don't Care

Exercise_01



Step 3: Flip-Flop Transition Table

Map between the each output transition and the inputs of JK Flip-Flop

Next-state table for 3-bit counter

Present State		Next State		Inputs of the JK-Flip Flop	
Q2		Q2		J2	K2
0		0		0	X
0		0		0	X
0		0		0	X
0		1		1	X
1		1		X	0
1		1		X	0
1		1		X	0
1		0		X	1

Transition table for a J-K flip-flop

Output Transistors			Flip-Flop Inputs	
Q_N		Q_{N+1}	J	K
0	→	0	0	X
0	→	1	1	X
1	→	0	X	1
1	→	1	X	0

Q_N : Present State

Q_{N+1} : Next State

X: Don't Care

Exercise_01



Step 3: Flip-Flop Transition Table

Map between the each output transition and the inputs of JK Flip-Flop



Next-state table for 3-bit counter

Present State			Next State			Inputs of the JK-Flip Flop					
Q2	Q1	Q0	Q2	Q1	Q0	J2	K2	J1	K1	J0	K0
0	0	0	0	0	1	0	X	0	X	1	X
0	0	1	0	1	0	0	X	1	X	X	1
0	1	0	0	1	1	0	X	X	0	1	X
0	1	1	1	0	0	1	X	X	1	X	1
1	0	0	1	0	1	X	0	0	X	1	X
1	0	1	1	1	0	X	0	1	X	X	1
1	1	0	1	1	1	X	0	X	0	1	X
1	1	1	0	0	0	X	1	X	1	X	1

Transition table for a J-K flip-flop

Output Transistors			Flip-Flop Inputs	
Q _N		Q _{N+1}	J	K
0	→	0	0	X
0	→	1	1	X
1	→	0	X	1
1	→	1	X	0

Q_N : Present State

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Exercise_01

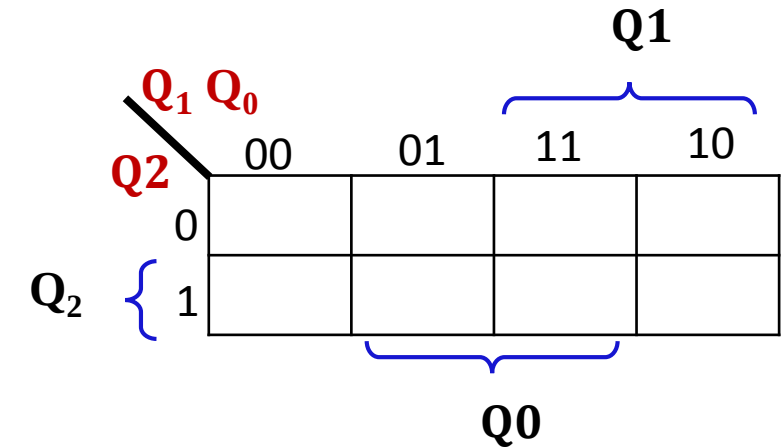


Step 4: k-Maps

K-maps can be used to determine the logic required for the J and K inputs of each flip-flop in the counter

Next-state table for 3-bit counter

Present State			Next State			Inputs of the JK-Flip Flop					
Q2	Q1	Q0	Q2	Q1	Q0	J2	K2	J1	K1	J0	K0
0	0	0	0	0	1	0	X	0	X	1	X
0	0	1	0	1	0	0	X	1	X	X	1
0	1	0	0	1	1	0	X	X	0	1	X
0	1	1	1	0	0	1	X	X	1	X	1
1	0	0	1	0	1	X	0	0	X	1	X
1	0	1	1	1	0	X	0	1	X	X	1
1	1	0	1	1	1	X	0	X	0	1	X
1	1	1	0	0	0	X	1	X	1	X	1



Exercise_01

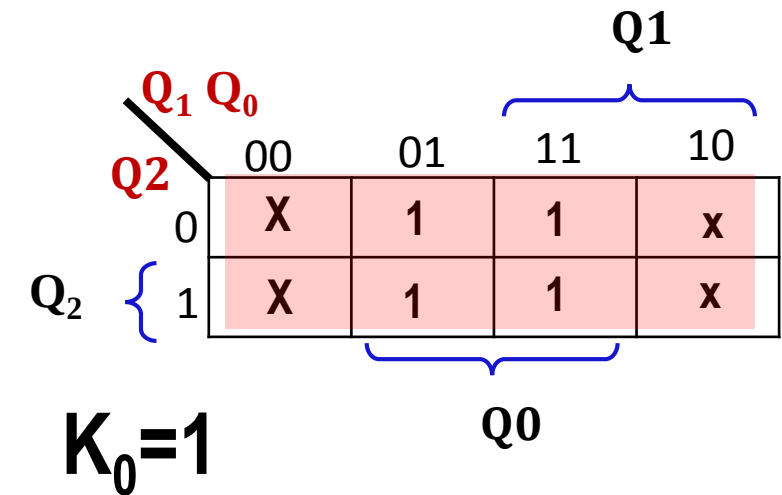


❑ Step 4: k-Maps

❑ K-maps can be used to determine the logic required for the J and K inputs of each flip-flop in the counter

Next-state table for 3-bit counter

Present State			Next State			Inputs of the JK-Flip Flop					
Q2	Q1	Q0	Q2	Q1	Q0	J2	K2	J1	K1	J0	K0
0	0	0	0	0	1	0	X	0	X	1	X
0	0	1	0	1	0	0	X	1	X	X	1
0	1	0	0	1	1	0	X	X	0	1	X
0	1	1	1	0	0	1	X	X	1	X	1
1	0	0	1	0	1	X	0	0	X	1	X
1	0	1	1	1	0	X	0	1	X	X	1
1	1	0	1	1	1	X	0	X	0	1	X
1	1	1	0	0	0	X	1	X	1	X	1



Exersice_01



❑ Step 4: k-Maps

❑ K-maps can be used to determine the logic required for the J and K inputs of each flip-flop in the counter

Next-state table for 3-bit counter

Present State			Next State			Inputs of the JK-Flip Flop					
Q2	Q1	Q0	Q2	Q1	Q0	J2	K2	J1	K1	J0	K0
0	0	0	0	0	1	0	X	0	X	1	X
0	0	1	0	1	0	0	X	1	X	X	1
0	1	0	0	1	1	0	X	X	0	1	X
0	1	1	1	0	0	1	X	X	1	X	1
1	0	0	1	0	1	X	0	0	X	1	X
1	0	1	1	1	0	X	0	1	X	X	1
1	1	0	1	1	1	X	0	X	0	1	X
1	1	1	0	0	0	X	1	X	1	X	1

Diagram illustrating a Karnaugh map (K-map) for a 3-variable function (likely a 3-bit adder or similar logic circuit). The map is a 2x4 grid.

The variables are labeled as Q_1 , Q_0 , and Q_2 .

The columns are labeled with the combinations of Q_1 and Q_0 : 00, 01, 11, 10.

The rows are labeled with the value of Q_2 : 0 and 1.

The cells contain the function values (X for unknown, 1 for true, x for don't care):

Q_2	$Q_1 Q_0$ = 00	$Q_1 Q_0$ = 01	$Q_1 Q_0$ = 11	$Q_1 Q_0$ = 10
0	X	1	1	x
1	X	1	1	x

Annotations:

- A blue bracket groups the columns 11 and 10.
- A blue bracket groups the columns 01 and 11.
- A red diagonal line separates the columns 00 from the others, labeled $Q_1 Q_0$.
- A blue bracket on the left groups the rows 0 and 1, labeled Q_2 .
- The bottom left corner is labeled $K-1$.

		Q1					
		Q1 Q0		00	01	11	10
Q2	0						
	1						

Exercise_01



Step 4: k-Maps

K-maps can be used to determine the logic required for the J and K inputs of each flip-flop in the counter

Next-state table for 3-bit counter

Present State			Next State			Inputs of the JK-Flip Flop					
Q2	Q1	Q0	Q2	Q1	Q0	J2	K2	J1	K1	J0	K0
0	0	0	0	0	1	0	X	0	X	1	X
0	0	1	0	1	0	0	X	1	X	X	1
0	1	0	0	1	1	0	X	X	0	1	X
0	1	1	1	0	0	1	X	X	1	X	1
1	0	0	1	0	1	X	0	0	X	1	X
1	0	1	1	1	0	X	0	1	X	X	1
1	1	0	1	1	1	X	0	X	0	1	X
1	1	1	0	0	0	X	1	X	1	X	1

		Q1			
		00	01	11	10
Q2	0	X	1	1	x
	1	X	1	1	x

$K_0=1$

		Q1			
		00	01	11	10
Q2	0	1	X	X	1
	1	1	X	X	1

$J_0=1$

Exersice_01



□ Step 4: k-Maps

□ K-maps can be used to determine the logic required for the J and K inputs of each flip-flop in the counter

Next-state table for 3-bit counter

Present State			Next State			Inputs of the JK-Flip Flop					
Q2	Q1	Q0	Q2	Q1	Q0	J2	K2	J1	K1	J0	K0
0	0	0	0	0	1	0	X	0	X	1	X
0	0	1	0	1	0	0	X	1	X	X	1
0	1	0	0	1	1	0	X	X	0	1	X
0	1	1	1	0	0	1	X	X	1	X	1
1	0	0	1	0	1	X	0	0	X	1	X
1	0	1	1	1	0	X	0	1	X	X	1
1	1	0	1	1	1	X	0	X	0	1	X
1	1	1	0	0	0	X	1	X	1	X	1

K₁ = Q₀

		Q ₁			
		00	01	11	10
Q ₂	0	X	X	1	0
	1	X	X	1	0

J₁ = Q₀

		Q ₁			
		00	01	11	10
Q ₂	0	0	1	X	X
	1	0	1	X	X

Exersice_01



□ Step 4: k-Maps

□ K-maps can be used to determine the logic required for the J and K inputs of each flip-flop in the counter

Next-state table for 3-bit counter

Present State			Next State			Inputs of the JK-Flip Flop					
Q2	Q1	Q0	Q2	Q1	Q0	J2	K2	J1	K1	J0	K0
0	0	0	0	0	1	0	X	0	X	1	X
0	0	1	0	1	0	0	X	1	X	X	1
0	1	0	0	1	1	0	X	X	0	1	X
0	1	1	1	0	0	1	X	X	1	X	1
1	0	0	1	0	1	X	0	0	X	1	X
1	0	1	1	1	0	X	0	1	X	X	1
1	1	0	1	1	1	X	0	X	0	1	X
1	1	1	0	0	0	X	1	X	1	X	1

K-map for K₂:

		Q ₁			
		Q ₀			
Q ₂	0	00	01	11	10
	1	0	0	1	0

$$K_2 = Q_0 \cdot Q_1$$

K-map for J₂:

		Q ₁			
		Q ₀			
Q ₂	0	00	01	11	10
	1	x	x	X	X

$$J_2 = Q_0 \cdot Q_1$$

Exercise_01



Step 5: Counter implementation

The final step is to implement the combinational logic from the expressions for the J and K inputs

$$J_0 = 1$$

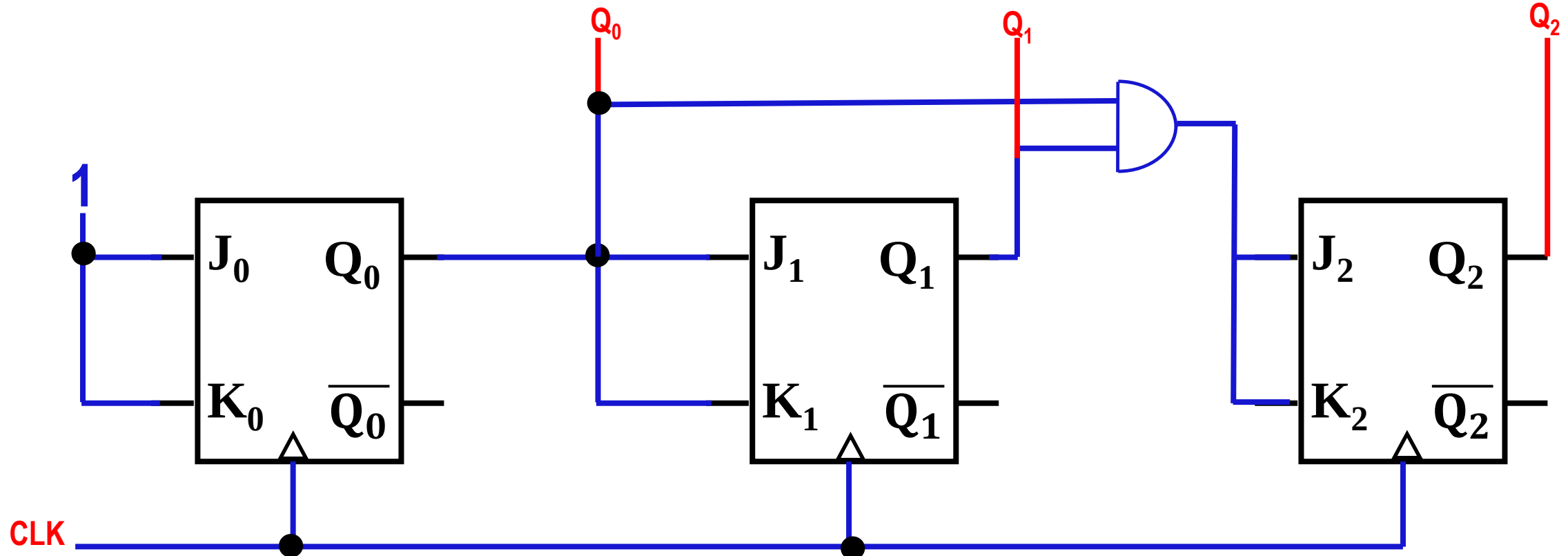
$$K_0 = 1$$

$$J_1 = Q_0$$

$$K_1 = Q_0$$

$$J_2 = Q_0 \cdot Q_1$$

$$K_2 = Q_0 \cdot Q_1$$





- ☐ Finite State Machine
- ☐ Asynchronous Counter
- ☐ Synchronous Counter
- ☐ Design of Synchronous Counters
- ☐ Exercise_01
- ☒ Exercise_02
- ☐ Exercise_03
- ☐ Exercise_04

Exercise_02

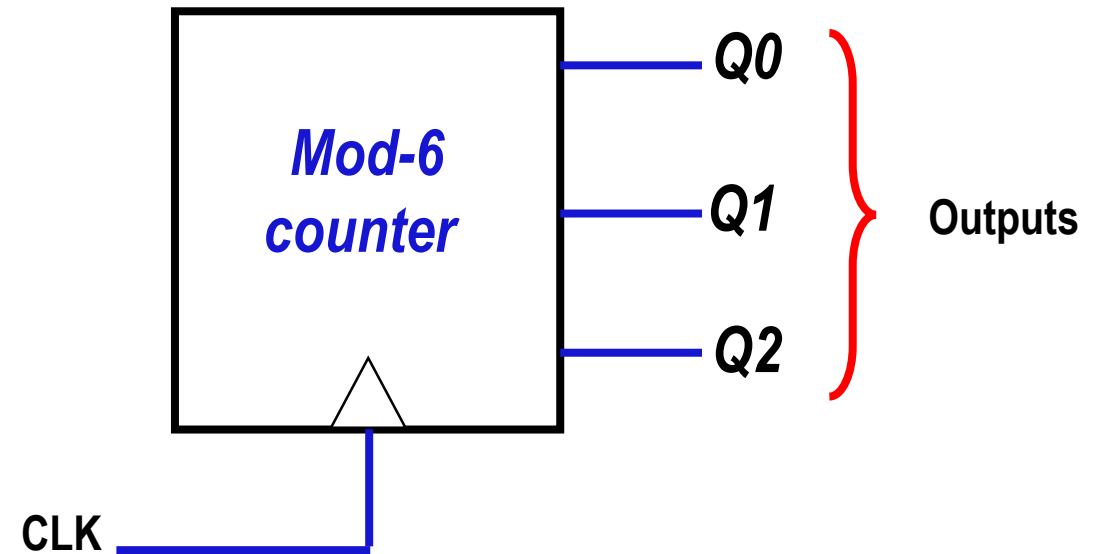


□ Design a **Mod-6** synchronous up Counter using JK flip flops.

Ans.

Mod-6 synchronous → 6 states

Mod-6 synchronous → needs 3 outputs

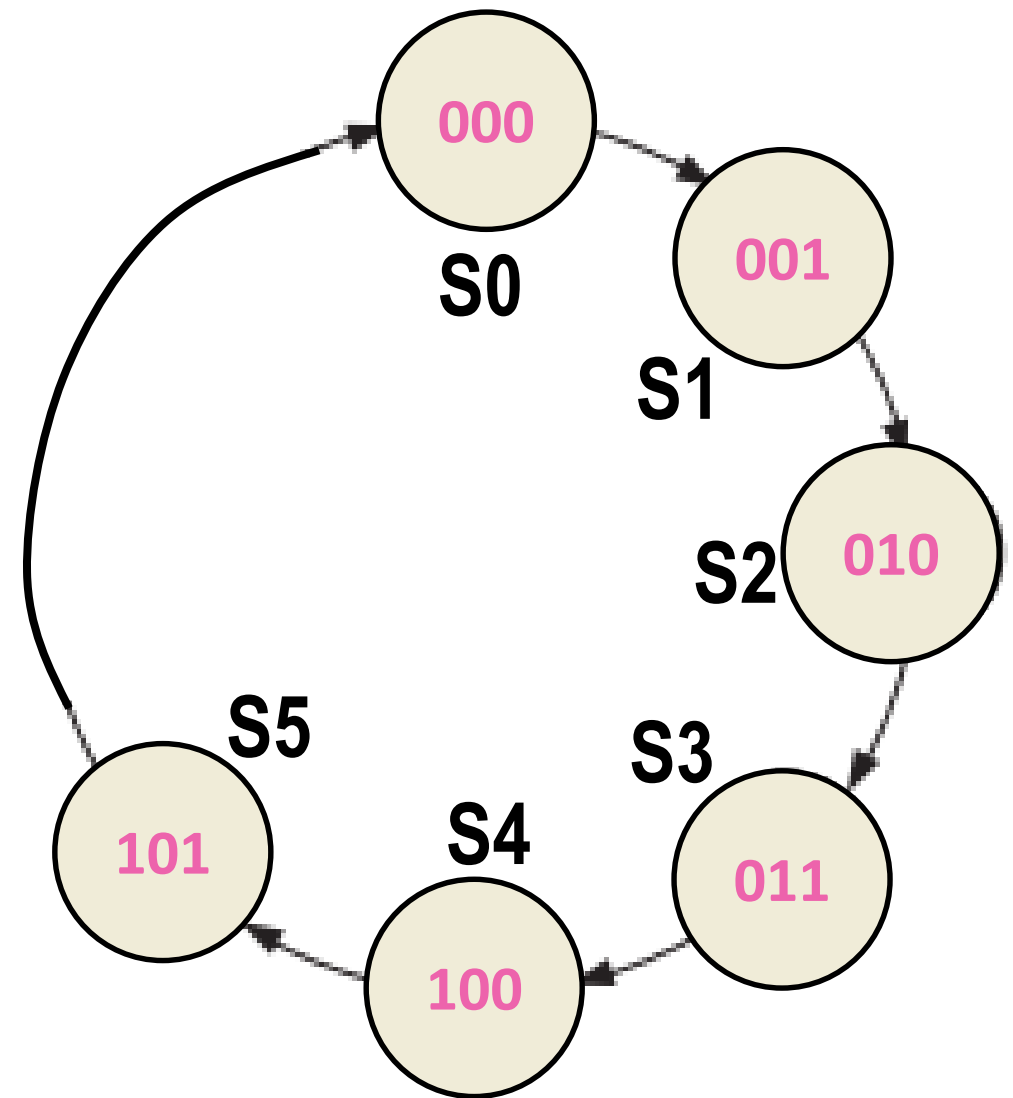
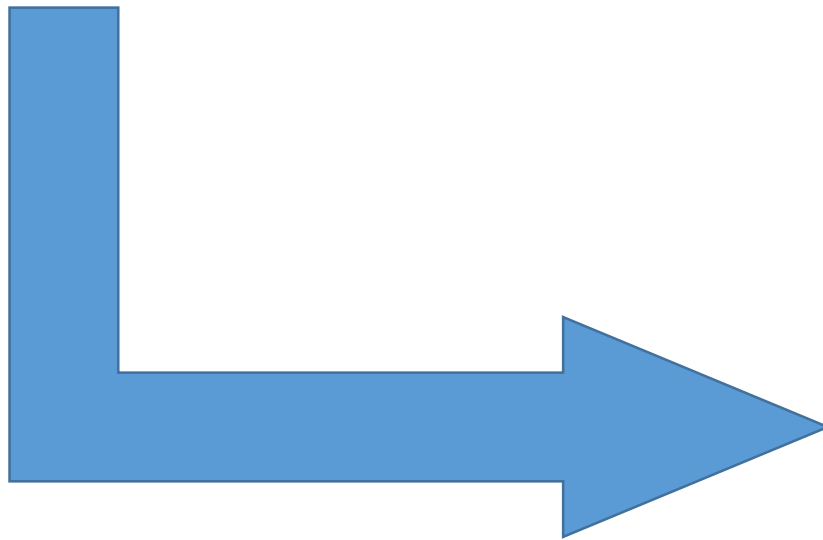


Exercise_02



□ Step 1: State Diagram

Mod-6 synchronous $\rightarrow$ 6 states
Up counter $\rightarrow \{0,1,2,3,4,5\}$



State Diagram
Mod-6 counter

Exercise_02



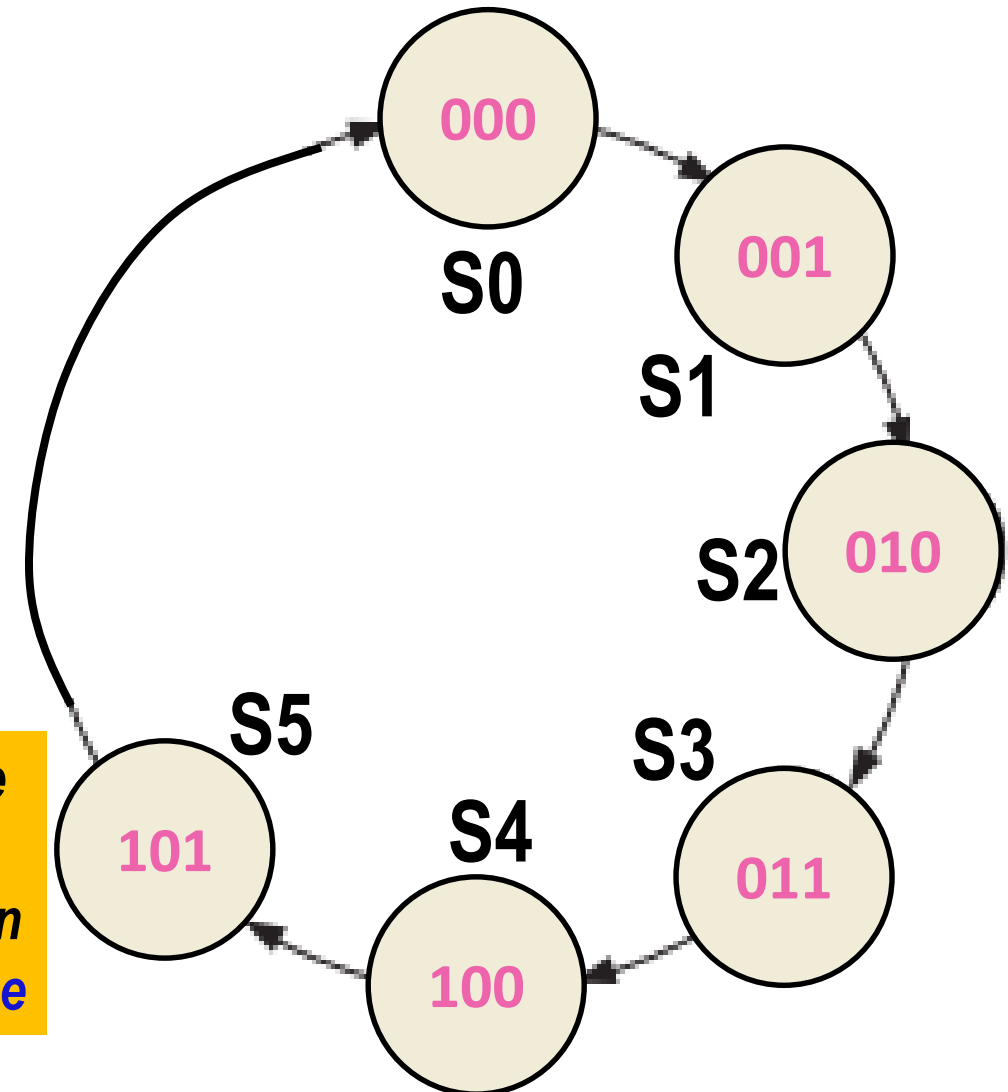
❑ Step 2: Next-State Table

- ❑ Once the sequential circuit is defined by a state diagram, the **second step** is to derive a next-state table

Next-state table for Mod-6 counter

Present State			Next State		
Q2	Q1	Q0	Q2	Q1	Q0

The **next state** is the state that the counter goes to from its present state upon application of a **clock pulse**



Exercise_02



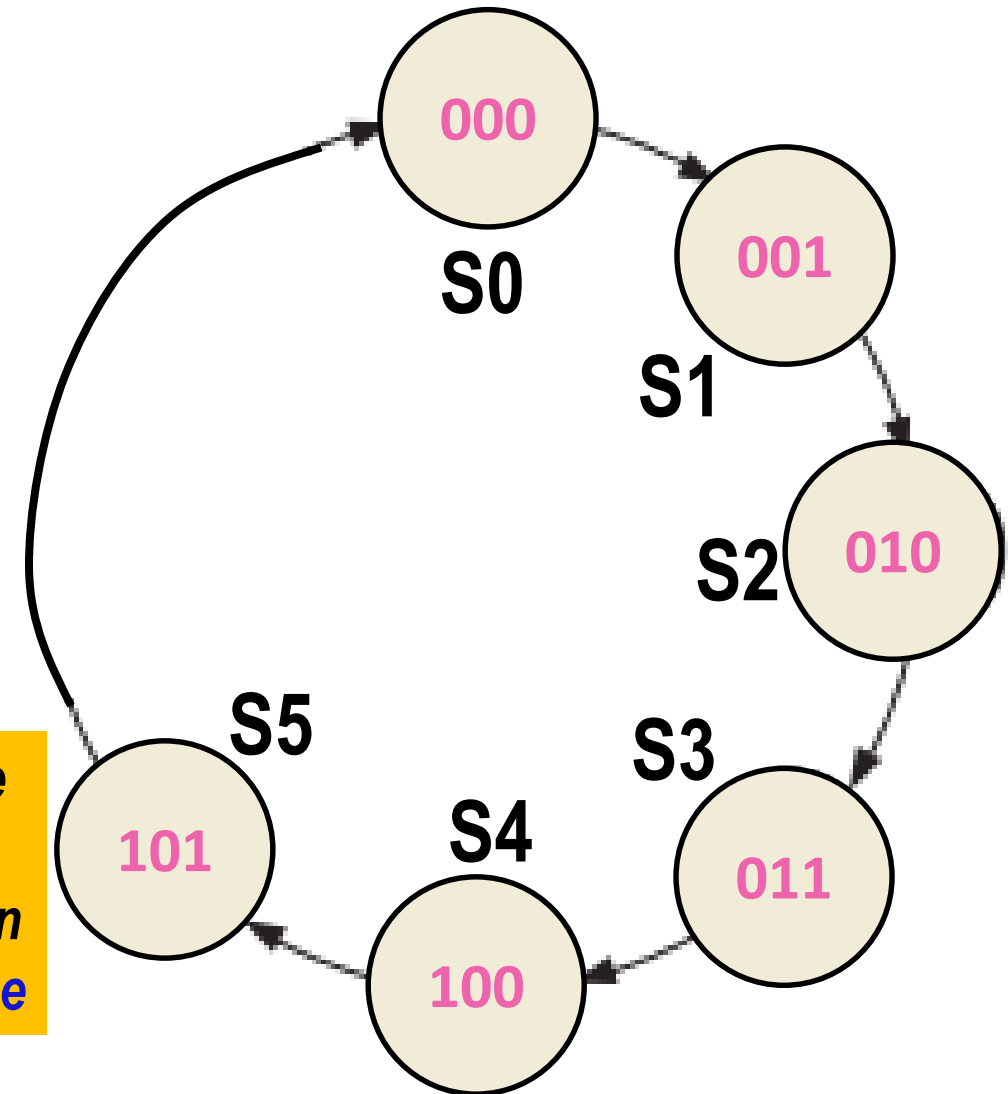
❑ Step 2: Next-State Table

❑ Once the sequential circuit is defined by a state diagram, the **second step** is to derive a next-state table

Next-state table for Mod-6 counter

Present State			Next State		
Q2	Q1	Q0	Q2	Q1	Q0
0	0	0	0	0	1
0	0	1	0	1	0
0	1	0	0	1	1
0	1	1	1	0	0
1	0	0	1	0	1
1	0	1	1	1	0
1	1	0	X	X	X
1	1	1	X	X	X

The **next state** is the state that the counter goes to from its present state upon application of a **clock pulse**



❑ Step 3: Flip-Flop Transition Table

❑ For each **output transition**, the J and K **inputs** that will cause the output transition to occur are always the same and defined in table.

Present State			Next State		
Q2	Q1	Q0	Q2	Q1	Q0
0	0	0	0	0	1
0	0	1	0	1	0
0	1	0	0	1	1
0	1	1	1	0	0
1	0	0	1	0	1
1	0	1	1	1	0
1	1	0	X	X	X
1	1	1	X	X	X

Transition table for a J-K flip-flop

Output Transistors			Flip-Flop Inputs	
Q_N		Q_{N+1}	J	K
0	→	0	0	X
0	→	1	1	X
1	→	0	X	1
1	→	1	X	0

Q_N : Present State

Q_{N+1} : Next State

X: Don't Care

Exercise_02



❑ Step 3: Flip-Flop Transition Table

❑ # of JK Flip-Flop = # of output of the counter. Here, # of JK Flip-Flop=3.

Next-state table for Mod-6 counter

Present State			Next State			Inputs of the JK-Flip Flop					
Q2	Q1	Q0	Q2	Q1	Q0	J2	K2	J1	K1	J0	K0
0	0	0	0	0	1						
0	0	1	0	1	0						
0	1	0	0	1	1						
0	1	1	1	0	0						
1	0	0	1	0	1						
1	0	1	0	0	0						
1	1	0	X	X	X						
1	1	1	X	X	X						

Transition table for a J-K flip-flop

Output Transistors			Flip-Flop Inputs	
Q _N		Q _{N+1}	J	K
0	→	0	0	X
0	→	1	1	X
1	→	0	X	1
1	→	1	X	0

Q_N : Present State

Q_{N+1} : Next State

X: Don't Care

Exersice_02

❑Step 3: Flip-Flop Transition Table

❑Map between the each output transition and the inputs of JK Flip-Flop

Next-state table for Mod-6 counter

Present State		Next State		Inputs of the JK-Flip Flop		
	Q0		Q0		J0	K0
	0		1			
	1		0			
	0		1			
	1		0			
	0		1			
	1		0			
	0		X			
	1		X			

Transition table for a J-K flip-flop

Output Transistors			Flip-Flop Inputs	
Q _N		Q _{N+1}	J	K
0	→	0	0	X
0	→	1	1	X
1	→	0	X	1
1	→	1	X	0

Q_N : Present State
 Q_{N+1} : Next State
 X: Don't Care

Exersice_02



❑ Step 3: Flip-Flop Transition Table

❑ Map between the each output transition and the inputs of JK Flip-Flop

Next-state table for Mod-6 counter

Present State		Next State		Inputs of the JK-Flip Flop	
	Q0		Q0	J0	K0
	0		1	1	X
	1		0	X	1
	0		1	1	X
	1		0	X	1
	0		1	1	X
	1		0	X	1
	0		X	X	X
	1		X	X	X

Transition table for a J-K flip-flop

Output Transistors			Flip-Flop Inputs	
Q_N		Q_{N+1}	J	K
0	→	0	0	X
0	→	1	1	X
1	→	0	X	1
1	→	1	X	0

Q_N : Present State

Q_{N+1} : Next State

X: Don't Care

Exersice_02

❑Step 3: Flip-Flop Transition Table

❑Map between the each output transition and the inputs of JK Flip-Flop

Next-state table for Mod-6 counter

Present State		Next State		Inputs of the JK-Flip Flop	
Q1		Q1		J1	K1
0		0			
0		1			
1		1			
1		0			
0		0			
0		0			
1		X			
1		X			

Transition table for a J-K flip-flop

Output Transistors			Flip-Flop Inputs	
Q_N		Q_{N+1}	J	K
0	→	0	0	X
0	→	1	1	X
1	→	0	X	1
1	→	1	X	0

Q_N : Present State
 Q_{N+1} : Next State
 X: Don't Care

Exercise_02



Step 3: Flip-Flop Transition Table

Map between the each output transition and the inputs of JK Flip-Flop

Next-state table for Mod-6 counter

Present State		Next State		Inputs of the JK-Flip Flop	
Q1		Q1		J1	K1
0		0		0	X
0		1		1	X
1		1		X	0
1		0		X	1
0		0		0	X
0		0		0	X
1		X		X	X
1		X		X	X

Transition table for a J-K flip-flop

Output Transistors			Flip-Flop Inputs	
Q_N		Q_{N+1}	J	K
0	→	0	0	X
0	→	1	1	X
1	→	0	X	1
1	→	1	X	0

Q_N : Present State

Q_{N+1} : Next State

X: Don't Care

Exersice_02

- ❑ Step 3: Flip-Flop Transition Table
 - ❑ Map between the each output transition and the inputs of JK Flip-Flop

Next-state table for Mod-6 counter

Present State			Next State		Inputs of the JK-Flip Flop		
Q2			Q2		J2	K2	
0			0				
0			0				
0			0				
0			1				
1			1				
1			0				
1			X				
1			X				

Transition table for a J-K flip-flop

Output Transistors			Flip-Flop Inputs	
Q _N		Q _{N+1}	J	K
0	→	0	0	X
0	→	1	1	X
1	→	0	X	1
1	→	1	X	0

Q_N : Present State
 Q_{N+1} : Next State
 X: Don't Care

Exercise_02



❑ Step 3: Flip-Flop Transition Table

❑ Map between the each output transition and the inputs of JK Flip-Flop

Next-state table for Mod-6 counter

Present State			Next State			Inputs of the JK-Flip Flop		
Q2			Q2			J2	K2	
0			0			0	X	
0			0			0	X	
0			0			0	X	
0			1			1	X	
1			1			X	0	
1			0			X	1	
1			X			X	X	
1			X			X	X	

Transition table for a J-K flip-flop

Output Transistors			Flip-Flop Inputs	
Q_N		Q_{N+1}	J	K
0	→	0	0	X
0	→	1	1	X
1	→	0	X	1
1	→	1	X	0

Q_N : Present State

Q_{N+1} : Next State

X: Don't Care

Step 3: Flip-Flop Transition Table

Map between the each output transition and the inputs of JK Flip-Flop

Next-state table for Mod-6 counter

Present State			Next State			Inputs of the JK-Flip Flop					
Q2	Q1	Q0	Q2	Q1	Q0	J2	K2	J1	K1	J0	K0
0	0	0	0	0	1	0	X	0	X	1	X
0	0	1	0	1	0	0	X	1	X	X	1
0	1	0	0	1	1	0	X	X	0	1	X
0	1	1	1	0	0	1	X	X	1	X	1
1	0	0	1	0	1	X	0	0	X	1	X
1	0	1	0	0	0	X	1	0	X	X	1
1	1	0	X	X	X	X	X	X	X	X	X
1	1	1	X	X	X	X	X	X	X	X	X

Transition table for a J-K flip-flop

Output Transistors			Flip-Flop Inputs	
Q_N		Q_{N+1}	J	K
0	→	0	0	X
0	→	1	1	X
1	→	0	X	1
1	→	1	X	0

Q_N : Present State

Q_{N+1} : Next State

X: Don't Care

Exersice_02

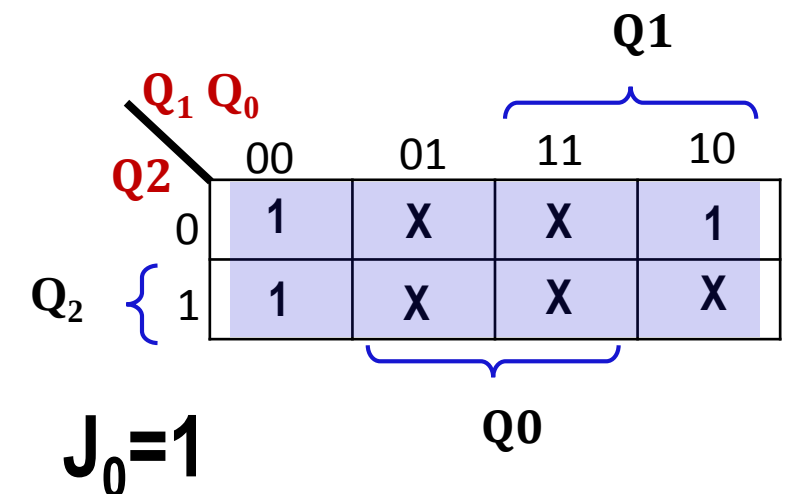
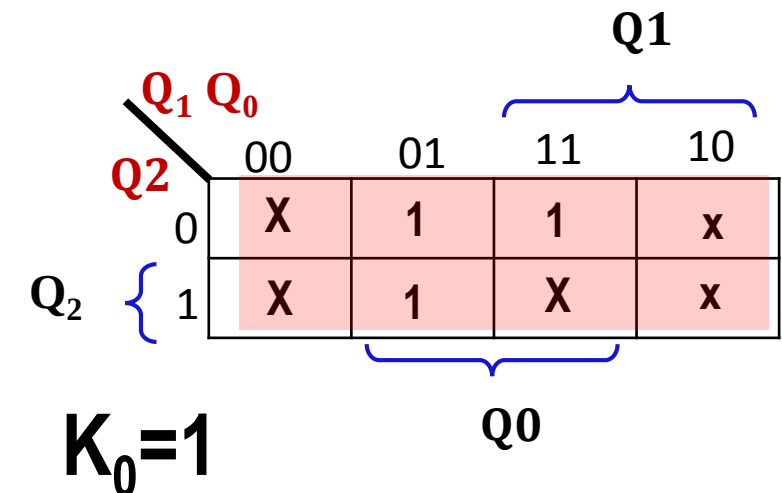


❑ Step 4: k-Maps

❑ K-maps can be used to determine the logic required for the J and K inputs of each flip-flop in the counter

Next-state table for Mod-6 counter

Present State			Next State			Inputs of the JK-Flip Flop					
Q2	Q1	Q0	Q2	Q1	Q0	J2	K2	J1	K1	J0	K0
0	0	0	0	0	1	0	X	0	X	1	X
0	0	1	0	1	0	0	X	1	X	X	1
0	1	0	0	1	1	0	X	X	0	1	X
0	1	1	1	0	0	1	X	X	1	X	1
1	0	0	1	0	1	X	0	0	X	1	X
1	0	1	0	0	0	X	1	0	X	X	1
1	1	0	X	X	X	X	X	X	X	X	X
1	1	1	X	X	X	X	X	X	X	X	X



Exercise_02



Step 4: k-Maps

K-maps can be used to determine the logic required for the J and K inputs of each flip-flop in the counter

Next-state table for Mod-6 counter

Present State			Next State			Inputs of the JK-Flip Flop					
Q2	Q1	Q0	Q2	Q1	Q0	J2	K2	J1	K1	J0	K0
0	0	0	0	0	1	0	X	0	X	1	X
0	0	1	0	1	0	0	X	1	X	X	1
0	1	0	0	1	1	0	X	X	0	1	X
0	1	1	1	0	0	1	X	X	1	X	1
1	0	0	1	0	1	X	0	0	X	1	X
1	0	1	0	0	0	X	1	0	X	X	1
1	1	0	X	X	X	X	X	X	X	X	X
1	1	1	X	X	X	X	X	X	X	X	X

Diagram illustrating a Karnaugh map for a 3-variable function (likely a 3-bit counter output) with variables Q_1 , Q_0 , and Q_2 .

The map is a 2x4 grid. The columns are labeled with $Q_1 Q_0$ pairs: 00, 01, 11, 10. The rows are labeled with Q_2 values: 0 and 1. The cells contain values: X, 1, 1, x for $Q_2=0$ and X, 1, X, x for $Q_2=1$.

Annotations include a blue bracket grouping the 11 and 10 columns under the label Q_1 , and another blue bracket grouping the 01 and 11 columns under the label Q_0 . A black diagonal line is drawn from the top-left corner (0,00) to the bottom-left corner (1,00).

$J_0=1$

		Q_1					
		$Q_1 Q_0$		00	01	11	10
Q_2	0	1	X	X	1		
	1	1	X	X	X		

Q_0

Exersice_02



❑ Step 4: k-Maps

❑ K-maps can be used to determine the logic required for the J and K inputs of each flip-flop in the counter

Next-state table for Mod-6 counter

Present State			Next State			Inputs of the JK-Flip Flop					
Q2	Q1	Q0	Q2	Q1	Q0	J2	K2	J1	K1	J0	K0
0	0	0	0	0	1	0	X	0	X	1	X
0	0	1	0	1	0	0	X	1	X	X	1
0	1	0	0	1	1	0	X	X	0	1	X
0	1	1	1	0	0	1	X	X	1	X	1
1	0	0	1	0	1	X	0	0	X	1	X
1	0	1	0	0	0	X	1	0	X	X	1
1	1	0	X	X	X	X	X	X	X	X	X
1	1	1	X	X	X	X	X	X	X	X	X

K₁ = Q₀

		Q ₁			
		00	01	11	10
Q ₂	0	X	X	1	0
	1	X	X	X	0

J₁ = Q₀ · $\overline{Q_2}$

		Q ₁			
		00	01	11	10
Q ₂	0	0	1	X	X
	1	0	0	X	X

Exercise_02



Step 4: k-Maps

K-maps can be used to determine the logic required for the J and K inputs of each flip-flop in the counter

Next-state table for Mod-6 counter

Present State			Next State			Inputs of the JK-Flip Flop					
Q2	Q1	Q0	Q2	Q1	Q0	J2	K2	J1	K1	J0	K0
0	0	0	0	0	1	0	X	0	X	1	X
0	0	1	0	1	0	0	X	1	X	X	1
0	1	0	0	1	1	0	X	X	0	1	X
0	1	1	1	0	0	1	X	X	1	X	1
1	0	0	1	0	1	X	0	0	X	1	X
1	0	1	0	0	0	X	1	0	X	X	1
1	1	0	X	X	X	X	X	X	X	X	X
1	1	1	X	X	X	X	X	X	X	X	X

K-map for K₂:

		Q ₁			
		00	01	11	10
Q ₂	0	X	X	X	X
	1	0	1	X	X

Grouping: K₂ = Q₀.

K-map for J₂:

		Q ₁			
		00	01	11	10
Q ₂	0	0	0	1	0
	1	x	x	X	X

Grouping: J₂ = Q₀ · Q₁.

Exersice_02



□ Step 5: Counter implementation

□ The final step is to implement the combinational logic from the expressions for the J and K inputs

$$J_0 = 1$$

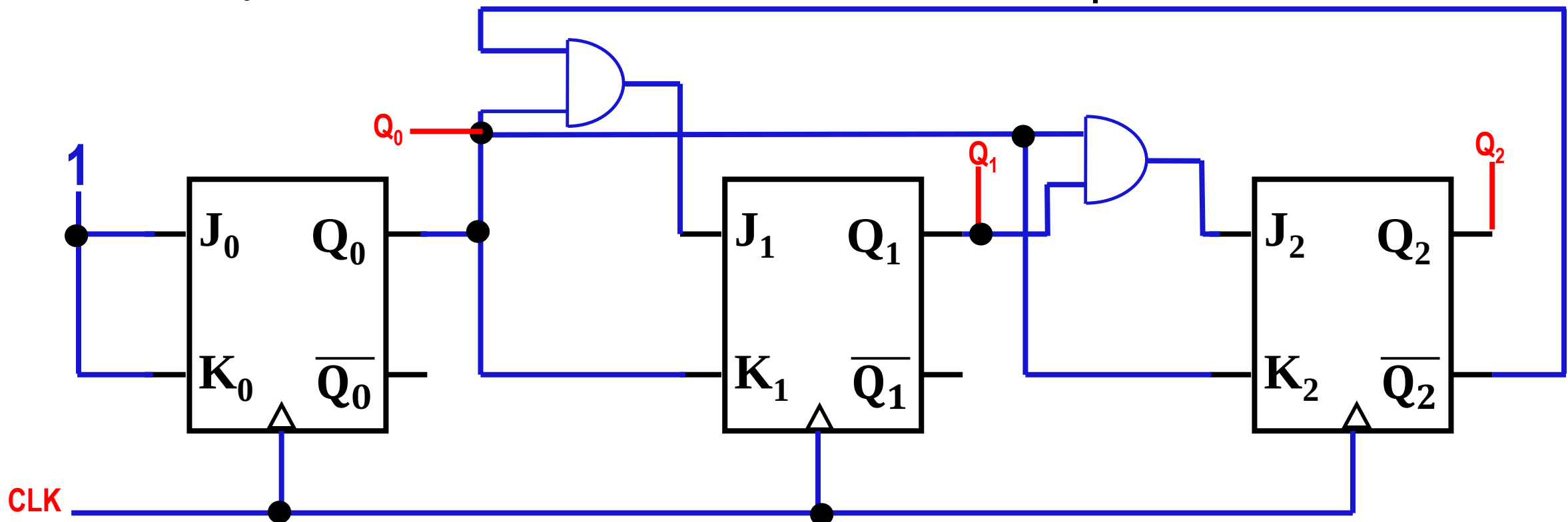
$$J_1 = Q_0 \cdot \overline{Q_2}$$

$$J_2 = Q_0 \cdot Q_1$$

$$K_0 = 1$$

$$K_1 = Q_0$$

$$K_2 = Q_0$$



- ☐ Finite State Machine
- ☐ Asynchronous Counter
- ☐ Synchronous Counter
- ☐ Design of Synchronous Counters
- ☐ Exercise_01
- ☐ Exercise_02
- ☒ Exercise_03
- ☐ Exercise_04

Exercise_03

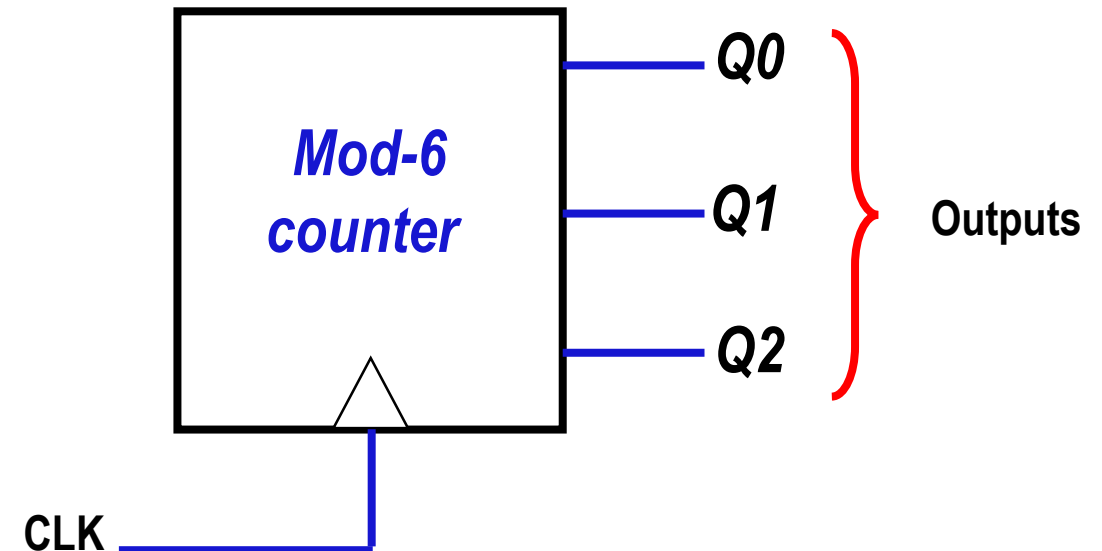


□ Design a synchronous counter with counting sequence: 000, 001, 011, 111, 110, 100, 000,...using JK flip flops.

Ans.

Mod-6 synchronous → 6 states

Mod-6 synchronous → needs 3 outputs

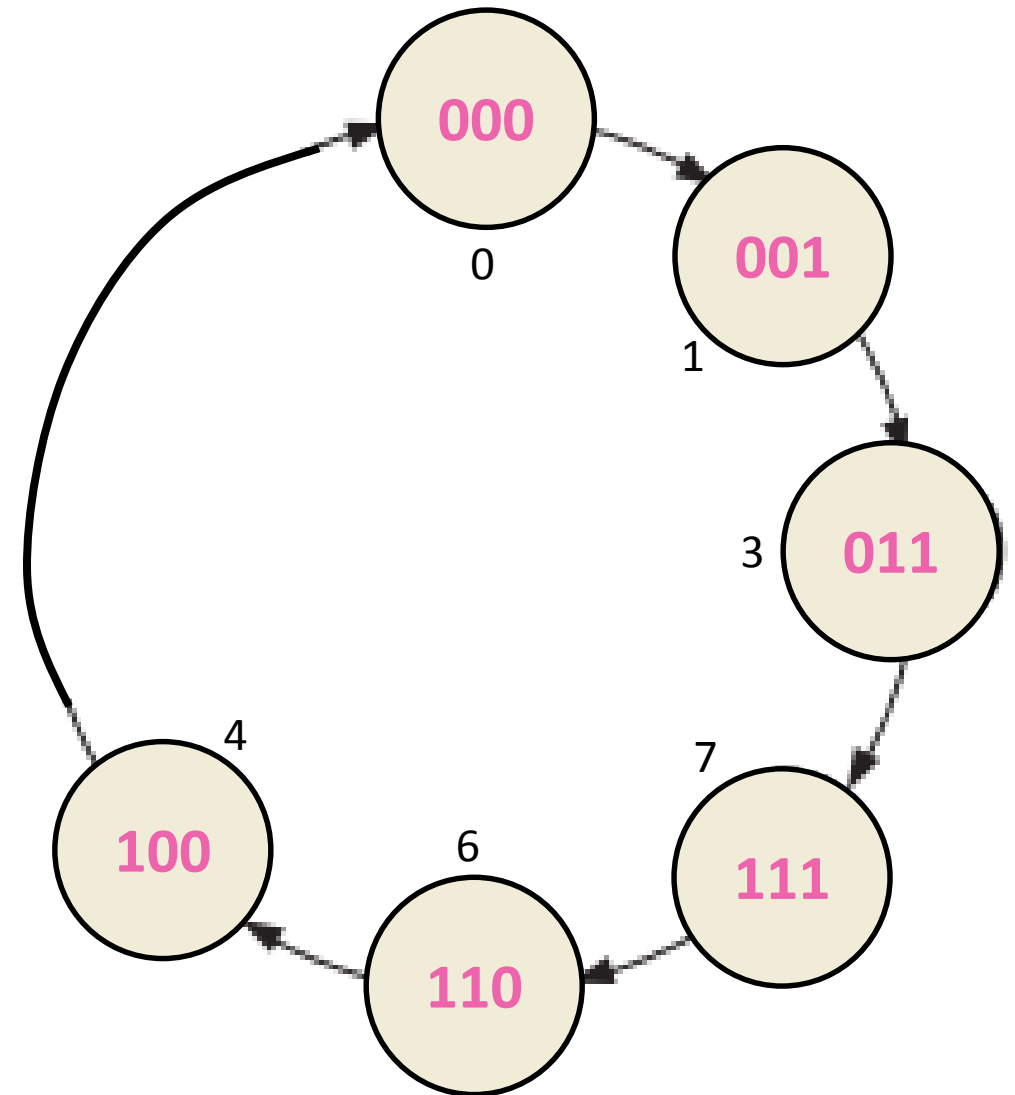
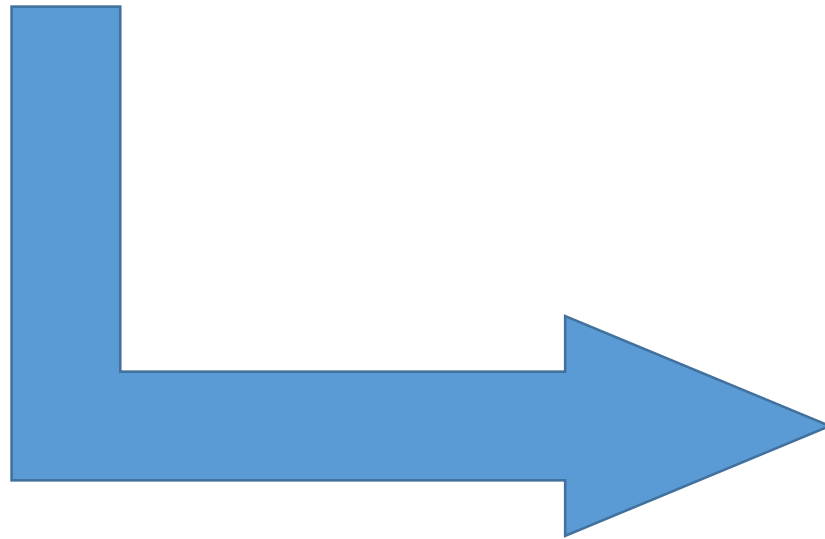


Exercise_03



□ Step 1: State Diagram

Mod-6 synchronous $\rightarrow$ 6 states
Up counter $\rightarrow \{0, 1, 3, 7, 6, 4\}$



State Diagram
Mod-6 counter

Exercise_03



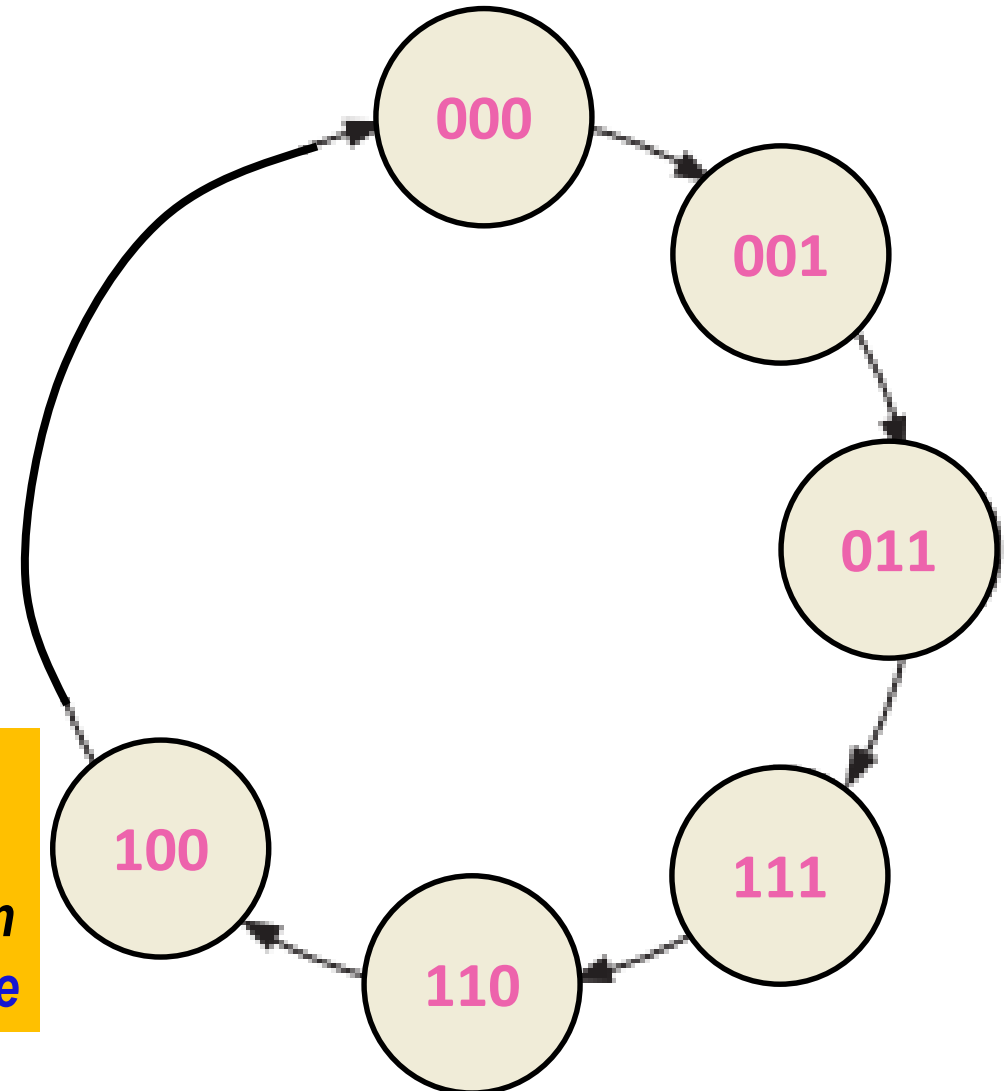
❑ Step 2: Next-State Table

- ❑ Once the sequential circuit is defined by a state diagram, the **second step** is to derive a next-state table

Next-state table

Present State			Next State		
Q2	Q1	Q0	Q2	Q1	Q0

The **next state** is the state that the counter goes to from its present state upon application of a **clock pulse**



Exersice_03



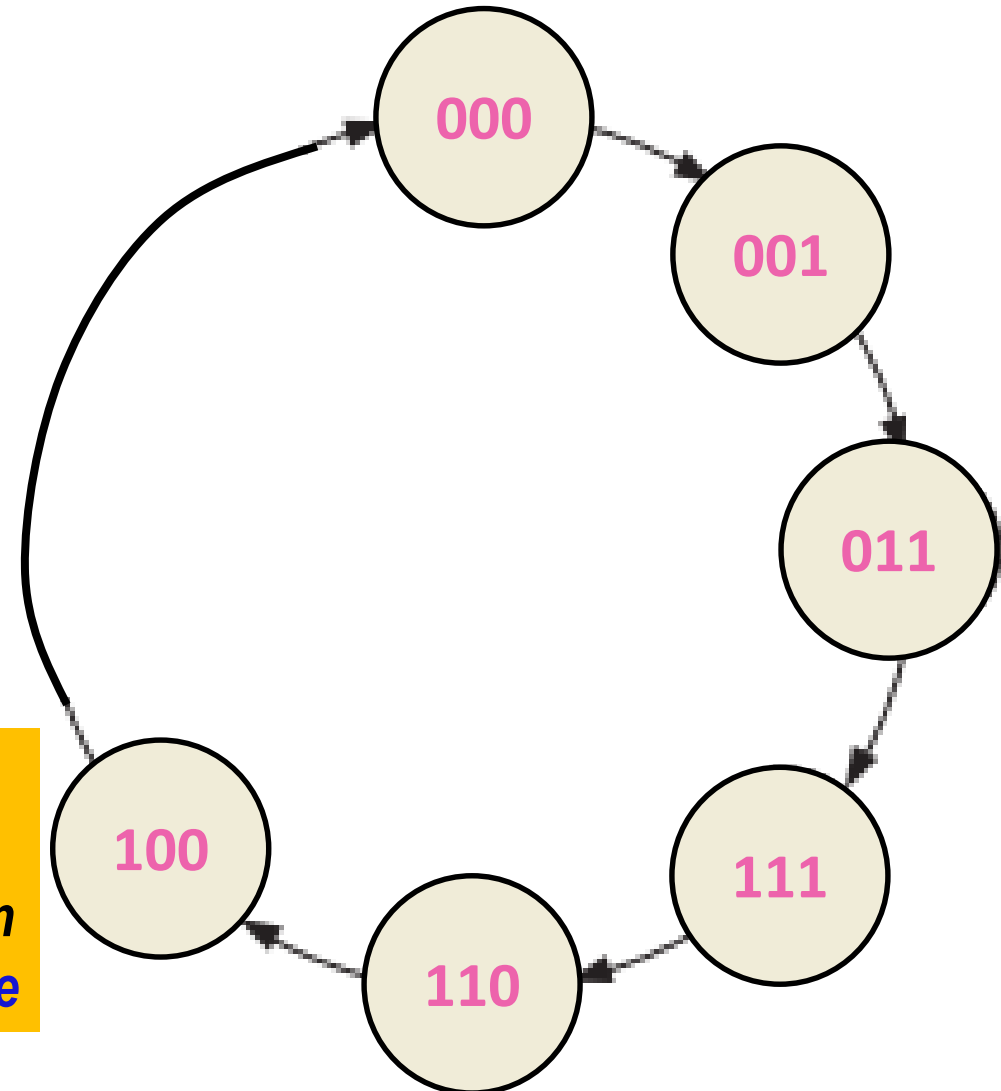
❑ Step 2: Next-State Table

❑ Once the sequential circuit is defined by a state diagram, the **second step** is to derive a next-state table

Next-state table

Present State			Next State		
Q2	Q1	Q0	Q2	Q1	Q0
0	0	0	0	0	1
0	0	1	0	1	1
0	1	0	X	X	X
0	1	1	1	1	1
1	0	0	0	0	0
1	0	1	X	X	X
1	1	0	0	0	0
1	1	1	1	1	0

The **next state** is the state that the counter goes to from its present state upon application of a **clock pulse**



❑ Step 3: Flip-Flop Transition Table

❑ For each **output transition**, the J and K **inputs** that will cause the output transition to occur are always the same and defined in table.

Next-state table

Present State			Next State		
Q2	Q1	Q0	Q2	Q1	Q0
0	0	0	0	0	1
0	0	1	0	1	1
0	1	0	X	X	X
0	1	1	1	1	1
1	0	0	0	0	0
1	0	1	X	X	X
1	1	0	0	0	0
1	1	1	1	1	0

Transition table for a J-K flip-flop

Output Transistors			Flip-Flop Inputs	
Q_N		Q_{N+1}	J	K
0	→	0	0	X
0	→	1	1	X
1	→	0	X	1
1	→	1	X	0

Q_N : Present State

Q_{N+1} : Next State

X: Don't Care

Exercise_03



❑ Step 3: Flip-Flop Transition Table

❑ # of JK Flip-Flop = # of output of the counter. Here, # of JK Flip-Flop=3.

Next-state table

Present State			Next State			Inputs of the JK-Flip Flop					
Q2	Q1	Q0	Q2	Q1	Q0	J2	K2	J1	K1	J0	K0
0	0	0	0	0	1						
0	0	1	0	1	1						
0	1	0	X	X	X						
0	1	1	1	1	1						
1	0	0	0	0	0						
1	0	1	X	X	X						
1	1	0	0	0	0						
1	1	1	1	1	0						

Transition table for a J-K flip-flop

Output Transistors			Flip-Flop Inputs	
Q _N		Q _{N+1}	J	K
0	→	0	0	X
0	→	1	1	X
1	→	0	X	1
1	→	1	X	0

Q_N : Present State

Q_{N+1} : Next State

X: Don't Care

Exersice_03



❑ Step 3: Flip-Flop Transition Table

❑ Map between the each output transition and the inputs of JK Flip-Flop

Next-state table

Present State		Next State		Inputs of the JK-Flip Flop	
	Q0		Q0	J0	K0
	0		1		
	1		1		
	0		X		
	1		1		
	0		0		
	1		X		
	0		0		
	1		0		

Transition table for a J-K flip-flop

Output Transistors			Flip-Flop Inputs	
Q_N		Q_{N+1}	J	K
0	→	0	0	X
0	→	1	1	X
1	→	0	X	1
1	→	1	X	0

Q_N : Present State

Q_{N+1} : Next State

X: Don't Care

Exersice_03



❑ Step 3: Flip-Flop Transition Table

❑ Map between the each output transition and the inputs of JK Flip-Flop

Next-state table

Present State		Next State		Inputs of the JK-Flip Flop	
	Q0		Q0	J0	K0
	0		1	1	X
	1		1	X	0
	0		X	X	X
	1		1	X	0
	0		0	0	X
	1		X	X	X
	0		0	0	X
	1		0	X	1

Transition table for a J-K flip-flop

Output Transistors			Flip-Flop Inputs	
Q_N		Q_{N+1}	J	K
0	→	0	0	X
0	→	1	1	X
1	→	0	X	1
1	→	1	X	0

Q_N : Present State

Q_{N+1} : Next State

X: Don't Care

Exercise_03



Step 3: Flip-Flop Transition Table

Map between the each output transition and the inputs of JK Flip-Flop

Next-state table

Present State		Next State		Inputs of the JK-Flip Flop	
	Q1		Q1	J1	K1
	0		0		
	0		1		
	1		X		
	1		1		
	0		0		
	0		X		
	1		0		
	1		1		

Transition table for a J-K flip-flop

Output Transistors			Flip-Flop Inputs	
Q_N		Q_{N+1}	J	K
0	→	0	0	X
0	→	1	1	X
1	→	0	X	1
1	→	1	X	0

Q_N : Present State

Q_{N+1} : Next State

X: Don't Care

Exercise_03



Step 3: Flip-Flop Transition Table

Map between the each output transition and the inputs of JK Flip-Flop

Next-state table

Present State		Next State		Inputs of the JK-Flip Flop	
	Q1		Q1	J1	K1
	0		0	0	X
	0		1	1	X
	1		X	X	X
	1		1	X	0
	0		0	0	X
	0		X	X	X
	1		0	X	1
	1		1	X	0

Transition table for a J-K flip-flop

Output Transistors			Flip-Flop Inputs	
Q_N		Q_{N+1}	J	K
0	→	0	0	X
0	→	1	1	X
1	→	0	X	1
1	→	1	X	0

Q_N : Present State

Q_{N+1} : Next State

X: Don't Care

Exercise_03



Step 3: Flip-Flop Transition Table

Map between the each output transition and the inputs of JK Flip-Flop

Next-state table

Present State			Next State			Inputs of the JK-Flip Flop		
Q2			Q2			J2	K2	
0			0					
0			0					
0			X					
0			1					
1			0					
1			X					
1			0					
1			1					

Transition table for a J-K flip-flop

Output Transistors			Flip-Flop Inputs	
Q_N		Q_{N+1}	J	K
0	→	0	0	X
0	→	1	1	X
1	→	0	X	1
1	→	1	X	0

Q_N : Present State

Q_{N+1} : Next State

X: Don't Care

Exercise_03



Step 3: Flip-Flop Transition Table

Map between the each output transition and the inputs of JK Flip-Flop

Next-state table

Present State			Next State			Inputs of the JK-Flip Flop		
Q2			Q2			J2	K2	
0			0			0	X	
0			0			0	X	
0			X			X	X	
0			1			1	X	
1			0			X	1	
1			X			X	X	
1			0			X	0	
1			1			X	0	

Transition table for a J-K flip-flop

Output Transistors			Flip-Flop Inputs	
Q_N		Q_{N+1}	J	K
0	→	0	0	X
0	→	1	1	X
1	→	0	X	1
1	→	1	X	0

Q_N : Present State

Q_{N+1} : Next State

X: Don't Care

Exercise_03



Step 3: Flip-Flop Transition Table

Map between the each output transition and the inputs of JK Flip-Flop

Next-state table

Present State			Next State			Inputs of the JK-Flip Flop					
Q2	Q1	Q0	Q2	Q1	Q0	J2	K2	J1	K1	J0	K0
0	0	0	0	0	1	0	X	0	X	1	X
0	0	1	0	1	1	0	X	1	X	X	0
0	1	0	X	X	X	X	X	X	X	X	X
0	1	1	1	1	1	1	X	X	0	X	0
1	0	0	0	0	0	X	1	0	X	0	X
1	0	1	X	X	X	X	X	X	X	X	X
1	1	0	0	0	0	X	0	X	1	0	X
1	1	1	1	1	0	X	0	X	0	X	1

Transition table for a J-K flip-flop

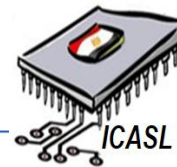
Output Transistors			Flip-Flop Inputs	
Q _N		Q _{N+1}	J	K
0	→	0	0	X
0	→	1	1	X
1	→	0	X	1
1	→	1	X	0

Q_N : Present State

Q_{N+1} : Next State

X: Don't Care

Exercise_03



Step 4: k-Maps

K-maps can be used to determine the logic required for the J and K inputs of each flip-flop in the counter

Next-state table

Present State			Next State			Inputs of the JK-Flip Flop					
Q2	Q1	Q0	Q2	Q1	Q0	J2	K2	J1	K1	J0	K0
0	0	0	0	0	1	0	X	0	X	1	X
0	0	1	0	1	1	0	X	1	X	X	0
0	1	0	X	X	X	X	X	X	X	X	X
0	1	1	1	1	1	1	X	X	0	X	0
1	0	0	0	0	0	X	1	0	X	0	X
1	0	1	X	X	X	X	X	X	X	X	X
1	1	0	0	0	0	X	0	X	1	0	X
1	1	1	1	1	0	X	0	X	0	X	1

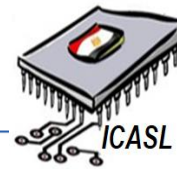
K₁ = $\overline{Q_0}$

		Q1			
		00	01	11	10
Q2	0	X	X	0	x
	1	X	X	0	1

J₁ = Q_0

		Q1			
		00	01	11	10
Q2	0	0	1	X	X
	1	0	X	X	X

Exercise_03



Step 4: k-Maps

K-maps can be used to determine the logic required for the J and K inputs of each flip-flop in the counter

Next-state table

Present State			Next State			Inputs of the JK-Flip Flop					
Q2	Q1	Q0	Q2	Q1	Q0	J2	K2	J1	K1	J0	K0
0	0	0	0	0	1	0	X	0	X	1	X
0	0	1	0	1	1	0	X	1	X	X	0
0	1	0	X	X	X	X	X	X	X	X	X
0	1	1	1	1	1	1	X	X	0	X	0
1	0	0	0	0	0	X	1	0	X	0	X
1	0	1	X	X	X	X	X	X	X	X	X
1	1	0	0	0	0	X	0	X	1	0	X
1	1	1	1	1	0	X	0	X	0	X	1

		Q1			
		Q0		Q0	
Q2	Q1 Q0	00	01	11	10
	0	X	X	X	x
1	1	1	X	0	0

$$K_2 = \overline{Q_1}$$

		Q1			
		Q0		Q0	
Q2	Q1 Q0	00	01	11	10
	0	0	0	1	X
1	1	X	X	X	X

$$J_2 = Q_1$$

Exersice_03



Step 5: Counter implementation

The final step is to implement the combinational logic from the expressions for the J and K inputs

$$J_0 = \overline{Q_2}$$

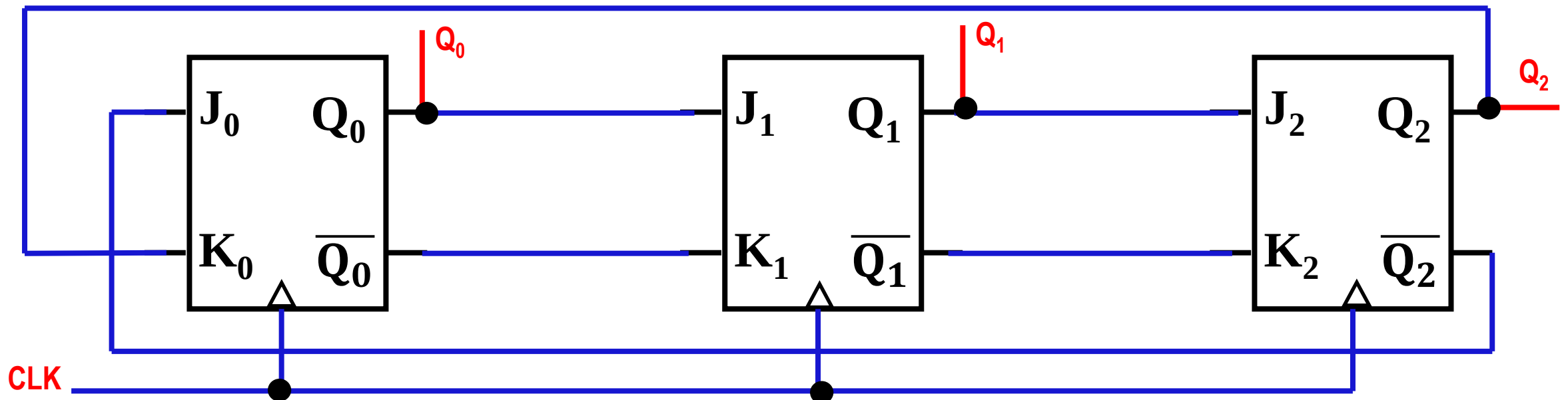
$$K_0 = Q_2$$

$$J_1 = Q_0$$

$$K_1 = \overline{Q_0}$$

$$J_2 = Q_1$$

$$K_2 = \overline{Q_1}$$





- ☐ Finite State Machine
- ☐ Asynchronous Counter
- ☐ Synchronous Counter
- ☐ Design of Synchronous Counters
- ☐ Exercise_01
- ☐ Exercise_02
- ☐ Exercise_03
- ☐ Exercise_04

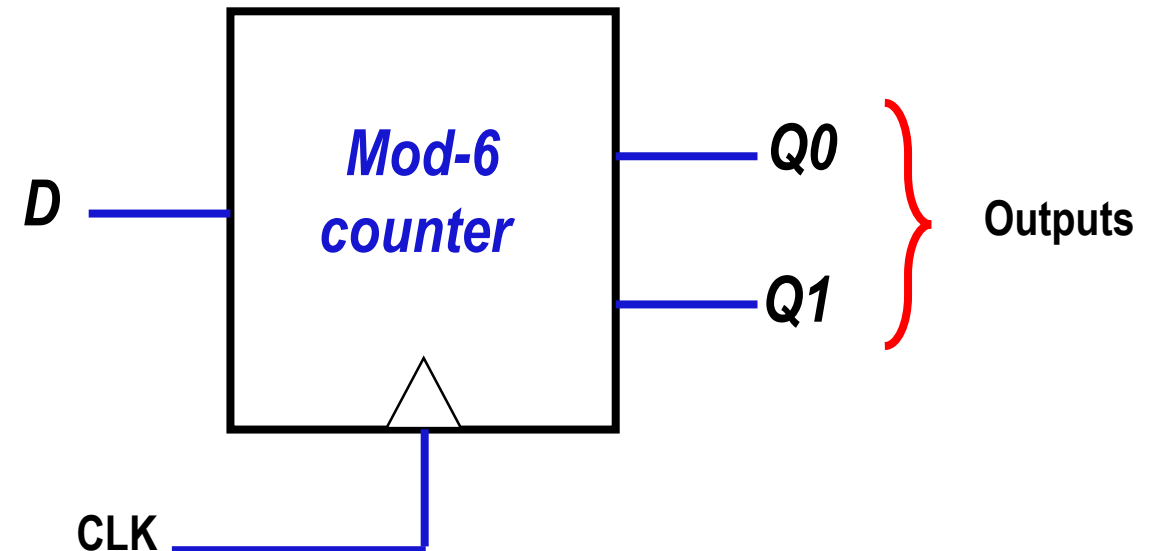
Exercise_04



□ Design a 2 bit up/down counter with an input D which determines the up/down function. Thus when $D=0$, the count sequence is 00,01,10,11,00 ... when $D=1$, the count sequence is 00,11,10,01,00 ...

Ans.

2 bit up/down counter $\rightarrow 2^2=4$ states

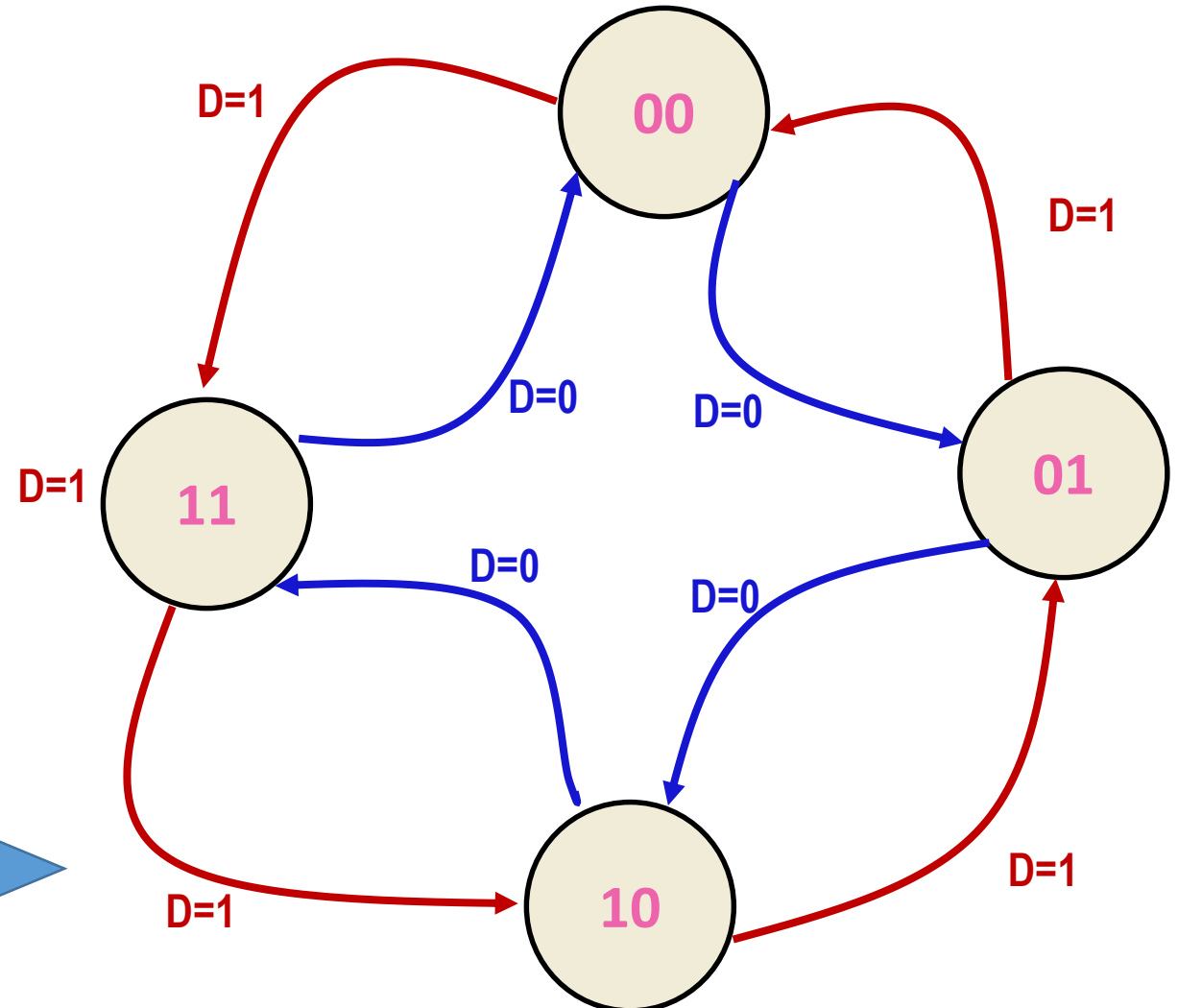


Exercise_04



□ Step 1: State Diagram

2-bit synchronous $\rightarrow 2^2 = 4$ states
Up counter $\rightarrow \{0,1,2,3\}$ @ $D=0$
Down counter $\rightarrow \{0,3,2,1\}$ @ $D=1$



State Diagram

Exercise_04



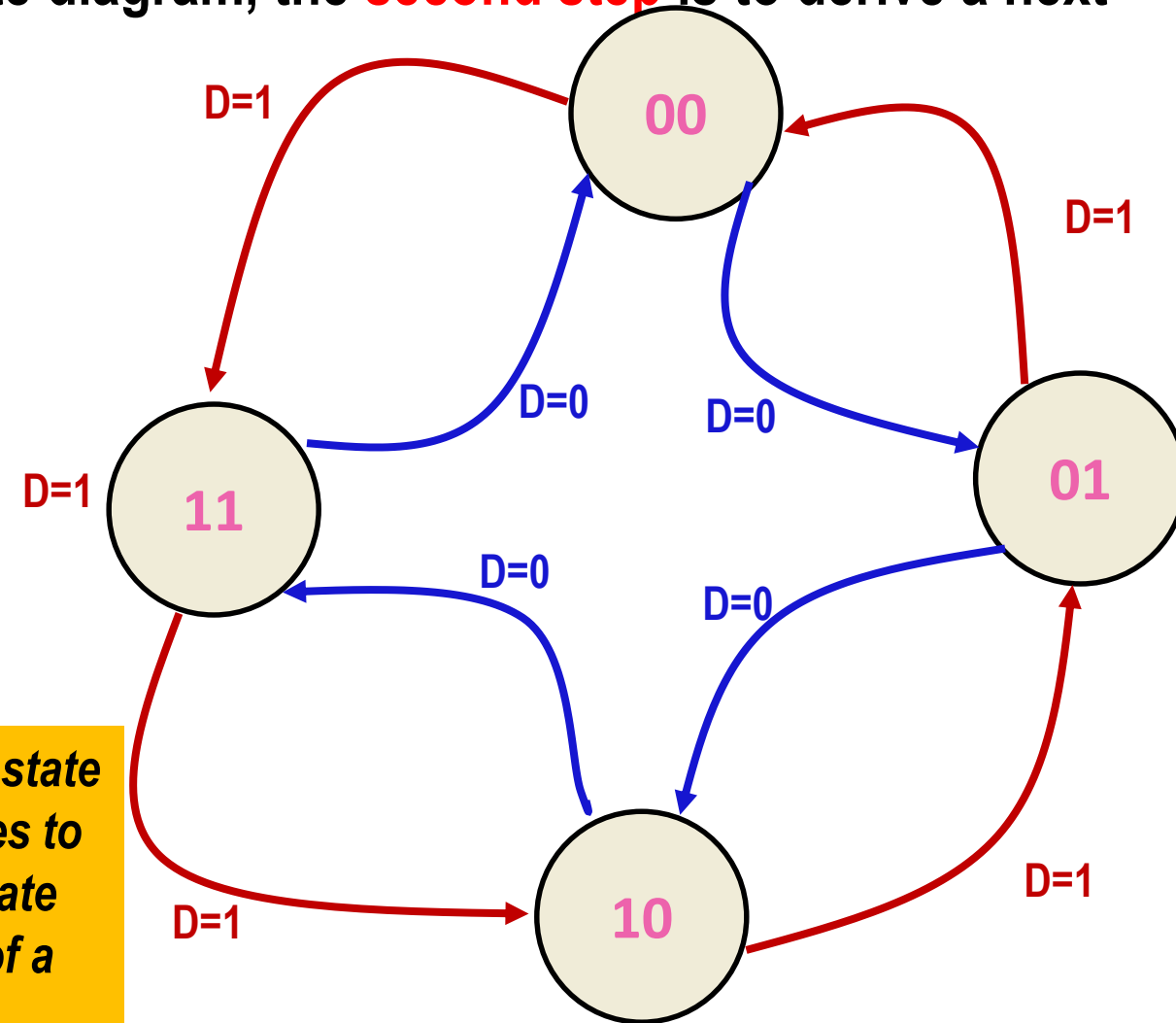
❑ Step 2: Next-State Table

❑ Once the sequential circuit is defined by a state diagram, the **second step** is to derive a next-state table

Next-state table

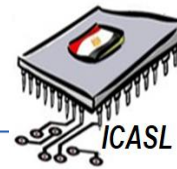
I/P	Present State		Next State	
D	Q1	Q0	Q1	Q0

The **next state** is the state that the counter goes to from its present state upon application of a **clock pulse**



State Diagram

Exercise_04



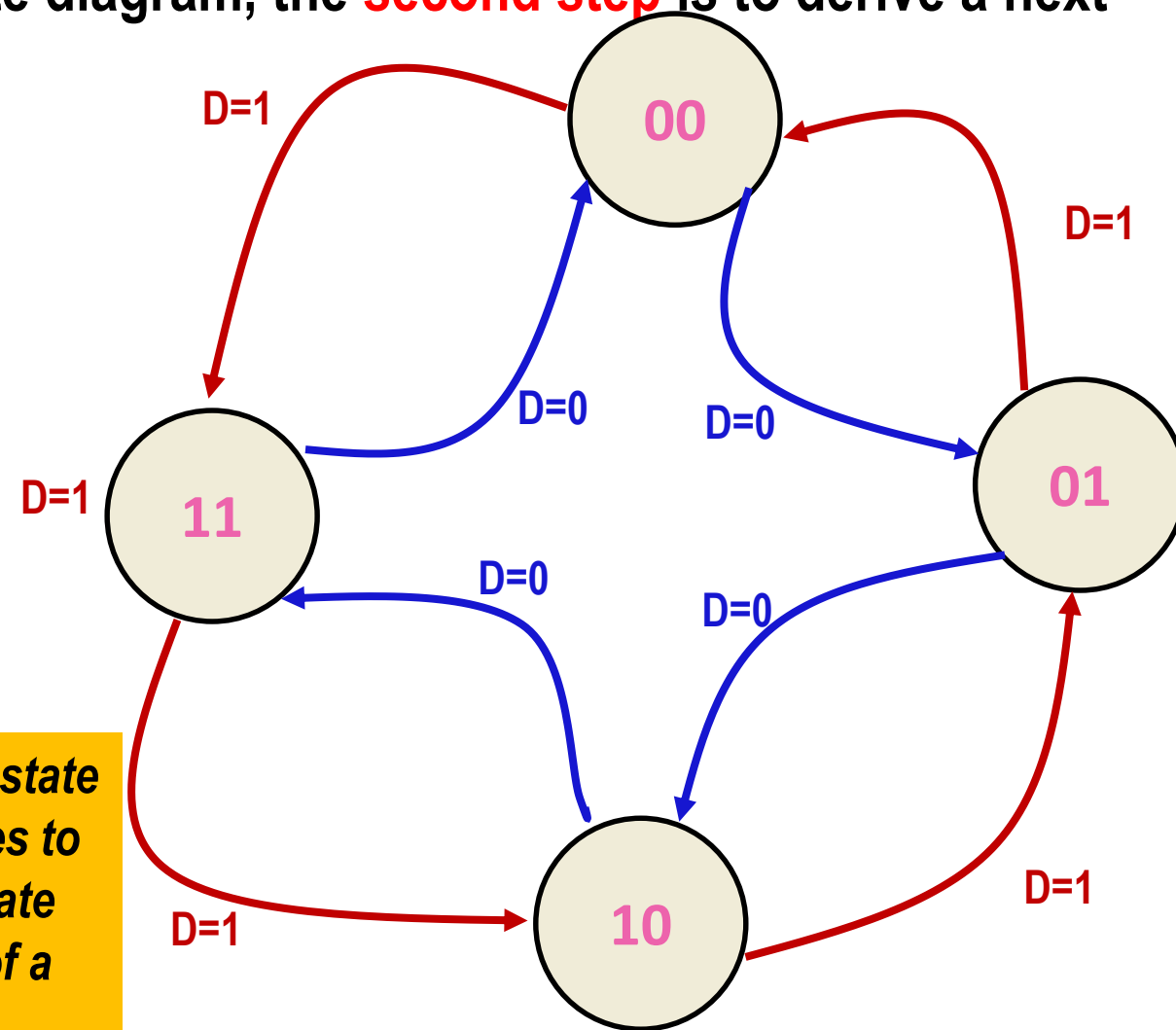
□ Step 2: Next-State Table

□ Once the sequential circuit is defined by a state diagram, the **second step** is to derive a next-state table

Next-state table

I/P	Present State		Next State	
D	Q1	Q0	Q1	Q0
0	0	0	0	1
0	0	1	1	0
0	1	0	1	1
0	1	1	0	0
1	0	0	1	1
1	0	1	0	0
1	1	0	0	1
1	1	1	1	0

The **next state** is the state that the counter goes to from its present state upon application of a **clock pulse**



State Diagram

❑ Step 3: Flip-Flop Transition Table

❑ For each **output transition**, the J and K **inputs** that will cause the output transition to occur are always the same and defined in table.

Next-state table

I/P	Present State		Next State	
D	Q1	Q0	Q1	Q0
0	0	0	0	1
0	0	1	1	0
0	1	0	1	1
0	1	1	0	0
1	0	0	1	1
1	0	1	0	0
1	1	0	0	1
1	1	1	1	0

Transition table for a J-K flip-flop

Output Transistors			Flip-Flop Inputs	
Q_N		Q_{N+1}	J	K
0	→	0	0	X
0	→	1	1	X
1	→	0	X	1
1	→	1	X	0

Q_N : Present State

Q_{N+1} : Next State

X: Don't Care

Exercise_04



❑ Step 3: Flip-Flop Transition Table

❑ # of JK Flip-Flop = # of output of the counter. Here, # of JK Flip-Flop=2.

Next-state table

I/P	Present State		Next State	
D	Q1	Q0	Q1	Q0
0	0	0	0	1
0	0	1	1	0
0	1	0	1	1
0	1	1	0	0
1	0	0	1	1
1	0	1	0	0
1	1	0	0	1
1	1	1	1	0

Inputs of the JK-Flip Flop			
J1	K1	J0	K0

Transition table for a J-K flip-flop

Output Transistors			Flip-Flop Inputs	
Q_N		Q_{N+1}	J	K
0	→	0	0	X
0	→	1	1	X
1	→	0	X	1
1	→	1	X	0

Q_N : Present State

Q_{N+1} : Next State

X: Don't Care

Exercise_04



Step 3: Flip-Flop Transition Table

Map between the each output transition and the inputs of JK Flip-Flop

Next-state table

I/P	Present State		Next State		Inputs of the JK-Flip Flop		
D		Q0		Q0		J0	K0
0		0		1			
0		1		0			
0		0		1			
0		1		0			
1		0		1			
1		1		0			
1		0		1			
1		1		0			

Transition table for a J-K flip-flop

Output Transistors			Flip-Flop Inputs	
Q_N		Q_{N+1}	J	K
0	→	0	0	X
0	→	1	1	X
1	→	0	X	1
1	→	1	X	0

Q_N : Present State

Q_{N+1} : Next State

X: Don't Care

Exercise_04



Step 3: Flip-Flop Transition Table

Map between the each output transition and the inputs of JK Flip-Flop

Next-state table

I/P	Present State		Next State		Inputs of the JK-Flip Flop		
D		Q0		Q0		J0	K0
0		0		1		1	X
0		1		0		X	1
0		0		1		1	X
0		1		0		X	1
1		0		1		1	X
1		1		0		X	1
1		0		1		1	X
1		1		0		X	1

Transition table for a J-K flip-flop

Output Transistors			Flip-Flop Inputs	
Q_N		Q_{N+1}	J	K
0	→	0	0	X
0	→	1	1	X
1	→	0	X	1
1	→	1	X	0

Q_N : Present State

Q_{N+1} : Next State

X: Don't Care

Step 3: Flip-Flop Transition Table

Map between the each output transition and the inputs of JK Flip-Flop

Next-state table

I/P	Present State		Next State		Inputs of the JK-Flip Flop		
D	Q1		Q1		J1	K1	
0	0		0				
0	0		1				
0	1		1				
0	1		0				
1	0		1				
1	0		0				
1	1		0				
1	1		1				

Transition table for a J-K flip-flop

Output Transistors			Flip-Flop Inputs	
Q_N		Q_{N+1}	J	K
0	→	0	0	X
0	→	1	1	X
1	→	0	X	1
1	→	1	X	0

Q_N : Present State

Q_{N+1} : Next State

X: Don't Care

Exercise_04



Step 3: Flip-Flop Transition Table

Map between the each output transition and the inputs of JK Flip-Flop

Next-state table

I/P	Present State		Next State		Inputs of the JK-Flip Flop		
D	Q1		Q1		J1	K1	
0	0		0		0	X	
0	0		1		1	X	
0	1		1		X	0	
0	1		0		X	1	
1	0		1		1	X	
1	0		0		0	X	
1	1		0		X	1	
1	1		1		X	0	

Transition table for a J-K flip-flop

Output Transistors			Flip-Flop Inputs	
Q_N		Q_{N+1}	J	K
0	→	0	0	X
0	→	1	1	X
1	→	0	X	1
1	→	1	X	0

Q_N : Present State

Q_{N+1} : Next State

X: Don't Care

Step 3: Flip-Flop Transition Table

Map between the each output transition and the inputs of JK Flip-Flop

Next-state table

I/P	Present State		Next State	
D	Q1	Q0	Q1	Q0
0	0	0	0	1
0	0	1	1	0
0	1	0	1	1
0	1	1	0	0
1	0	0	1	1
1	0	1	0	0
1	1	0	0	1
1	1	1	1	0

Inputs of the JK-Flip Flop			
J1	K1	J0	K0
0	X	1	X
1	X	X	1
X	0	1	X
X	1	X	1
1	X	1	X
0	X	X	1
X	1	1	X
X	0	X	1

Transition table for a J-K flip-flop

Output Transistors			Flip-Flop Inputs	
Q_N		Q_{N+1}	J	K
0	→	0	0	X
0	→	1	1	X
1	→	0	X	1
1	→	1	X	0

Q_N : Present State

Q_{N+1} : Next State

X: Don't Care

Exercise_04



Step 4: k-Maps

K-maps can be used to determine the logic required for the J and K inputs of each flip-flop in the counter

Next-state table

I/P	Present State		Next State		Inputs of the JK-Flip Flop			
	Q1	Q0	Q1	Q0	J1	K1	J0	K0
0	0	0	0	1	0	X	1	X
0	0	1	1	0	1	X	X	1
0	1	0	1	1	X	0	1	X
0	1	1	0	0	X	1	X	1
1	0	0	1	1	1	X	1	X
1	0	1	0	0	0	X	X	1
1	1	0	0	1	X	1	1	X
1	1	1	1	0	X	0	X	1

K₀=1

		Q1			
		00	01	11	10
D	0	X	1	1	x
	1	X	1	1	x

J₀=1

		Q1			
		00	01	11	10
D	0	1	X	X	1
	1	1	X	X	1

Exercise_04



Step 4: k-Maps

K-maps can be used to determine the logic required for the J and K inputs of each flip-flop in the counter

Next-state table

I/P	Present State		Next State	
D	Q1	Q0	Q1	Q0
0	0	0	0	1
0	0	1	1	0
0	1	0	1	1
0	1	1	0	0
1	0	0	1	1
1	0	1	0	0
1	1	0	0	1
1	1	1	1	0

Inputs of the JK-Flip Flop			
J1	K1	J0	K0
0	X	1	X
1	X	X	1
X	0	1	X
X	1	X	1
1	X	1	X
0	X	X	1
X	1	1	X
X	0	X	1

		Q1			
		00	01	11	10
D	0	X	X	1	0
	1	X	X	0	1

$$K_0 = \overline{D} \cdot Q_0 + D \cdot \overline{Q_0}$$

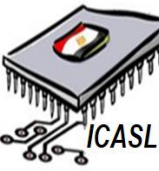
$$= Q_0 \oplus D$$

		Q1			
		00	01	11	10
D	0	0	1	X	X
	1	1	0	X	X

$$J_0 = \overline{D} \cdot Q_0 + D \cdot \overline{Q_0}$$

$$= Q_0 \oplus D$$

Exersice_04



□ Step 5: Counter implementation

□ The final step is to implement the combinational logic from the expressions for the J and K inputs

$$J_0=1$$

$$K_0=1$$

$$J_1=Q_0 \oplus D$$

$$K_1=Q_0 \oplus D$$

